



*Identifying Candidate Biomarkers of Psoriasis Through Convergence
Across Differential Expression Analysis, Cross-Validation Selection Stability,
and Machine Learning Feature Importance*

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Abstract

Background: Psoriasis is a common, chronic skin disease driven by immune system dysfunction and characterised by widespread dysregulation of gene activity. Machine learning is increasingly applied to psoriatic gene expression data, yet most studies restrict the search to a pre-defined category of genes and interpret at most a single model, rarely testing whether different algorithms agree on the genes driving their predictions.

Aims: This study aimed to identify robust candidate gene biomarkers of psoriasis through a genome-wide search requiring convergence across three lines of evidence: differential expression analysis, cross-validation selection stability, and machine learning feature importance, and to characterise their wider disease associations.

Methods: Using the public GSE54456 RNA-sequencing dataset (92 psoriatic and 82 non-psoriatic skin samples; 21,510 genes), differential expression analysis was conducted with limma. Within a leakage-controlled nested cross-validation framework, an elastic net performed genome-wide feature selection, and the final 64-gene panel was used to train three classifiers from different algorithm families (XGBoost, random forest, and a support vector machine), whose predictions were interpreted with SHAP. Genes converging across all three lines of evidence were retained as candidates and annotated for disease associations using DISGENET.

Results: Differential expression analysis identified 1,131 significantly dysregulated genes, recovering the established psoriasis signature. All three classifiers separated psoriatic from non-psoriatic skin without error, yet their feature-importance rankings diverged sharply, with *ABCG4* the only gene ranked highly by all three. Combining the three evidence lines yielded 14 candidate biomarkers: eight with established psoriasis associations and six with none. The most notable of the novel candidates was *ABCG4*, whose strongest recorded associations lie in prostate cancer and Alzheimer's disease rather than any skin condition.

Conclusion: Convergent, genome-wide analysis surfaced 14 candidate biomarkers, including novel genes, that category-restricted, single-model pipelines could not. These candidates are well supported within the GSE54456 cohort but require validation in independent cohorts and laboratory testing to establish any causal role. The study's main contribution is a transferable framework for prioritising biomarkers from high-dimensional expression data.

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Glossary of Terms and Abbreviations

This glossary defines the medical, statistical, and machine learning terms, acronyms, and gene symbols used in this dissertation. Gene symbols are shown in italics by convention.

1. Abbreviations and Acronyms

Abbreviation	Meaning
AI	Artificial intelligence.
AUC	Area Under the (ROC) Curve; a threshold-independent measure of classification performance.
CD8+ T cell	A type of cytotoxic (killer) immune cell; the gene panel of He et al. (2024) was derived from these.
COPD	Chronic Obstructive Pulmonary Disease.
CSV	Comma-Separated Values; a plain-text tabular file format.
CV	Cross-Validation.
DALEX	Descriptive Machine Learning Explanations; a model-explanation method.
DE	Differential Expression.
DSI	Disease Specificity Index; a DISGENET measure of how many diseases a gene is associated with.
F1	F1 score; the harmonic mean of precision and recall.
FC (log₂FC)	Fold Change; log ₂ FC is the fold change expressed on a log ₂ scale.
FDR	False Discovery Rate.
FN / FP / TN / TP	False Negative / False Positive / True Negative / True Positive.
GDA	Gene–Disease Association (score); a DISGENET score (0–1) for the strength of a gene–disease link.
GEO	Gene Expression Omnibus; NCBI’s public repository of gene expression datasets.
GSE54456	The GEO accession number of the psoriasis RNA-sequencing dataset used in this study.
hBD-2	Human β -defensin-2; the antimicrobial protein encoded by DEFB4A.
HGNC	HUGO Gene Nomenclature Committee; the authority for official gene symbols.
IL-8	Interleukin-8; the protein encoded by CXCL8.
LASSO	Least Absolute Shrinkage and Selection Operator; an L1-penalised regression (feature selection) method.
LIME	Local Interpretable Model-agnostic Explanations; a model-explanation method.
MCC	Matthews Correlation Coefficient; a balanced classification metric ranging from -1 to $+1$.
mtry	The number of features randomly considered at each split in a random forest.
NCBI	National Center for Biotechnology Information.
PCA (PC1, PC2)	Principal Component Analysis; PC1/PC2 are the first and second principal components.
PMID	PubMed Identifier; a unique reference number for a publication indexed in PubMed.
PTSD	Post-Traumatic Stress Disorder.
RF	Random Forest.
RNA / RNA-seq	Ribonucleic Acid / RNA sequencing.
ROC	Receiver Operating Characteristic (curve).

Abbreviation	Meaning
RPKM	Reads Per Kilobase per Million mapped reads; a normalised measure of gene expression level from RNA sequencing.
RQ1–RQ3	Research Questions 1 to 3.
SHAP	Shapley Additive Explanations; the method used to attribute each model’s predictions to individual genes.
SVM	Support Vector Machine.
SVM-RFE	Support Vector Machine Recursive Feature Elimination; a feature selection method.
XGB / XGBoost	Extreme Gradient Boosting; a gradient-boosted decision-tree algorithm.

2. Medical and Biological Terms

Term	Definition
Antimicrobial peptide	A small protein of the innate immune system that kills microbes and can recruit immune cells (e.g. hBD-2).
Biomarker	A measurable biological indicator (here, a gene) associated with a disease.
Candidate biomarker	A gene nominated by this study for further validation, not a confirmed biomarker.
Comorbidity	An additional disease occurring alongside the primary condition (e.g. cardiovascular disease with psoriasis).
Dysregulation	Abnormal gene expression; either increased or decreased relative to normal.
Epidermis / epidermal	The outermost layer of the skin; “epidermal hyperproliferation” is the excessive multiplication of keratinocytes seen in psoriasis.
Gene expression	The extent to which a gene is transcribed into RNA (its “activity”).
Gene / protein	A gene is a stretch of DNA that encodes a protein, the functional molecule it produces.
Immune dysfunction	Abnormal immune-system activity; in psoriasis an overactive response that targets the body’s own cells.
Inflammation / inflammatory cascade	The immune response that, in psoriasis, becomes a self-sustaining feedback loop driving the disease.
Keratinocyte	The predominant cell type of the epidermis (outer skin layer).
Lesion / lesional skin	Affected (diseased) skin; “lesional” contrasts with normal/non-lesional skin.
Pathogenesis	The biological process by which a disease develops.
Plaque	A raised, scaly, inflamed patch of skin characteristic of psoriasis.
Pleiotropy	The association of a single gene with several distinct diseases or traits.
Psoriasis	A chronic, inflammatory skin disease driven by immune-system dysfunction, characterised by scaly plaques.
Skin biopsy	A tissue sample of skin, from which gene expression was measured.
Squamous cell carcinoma	A form of cancer arising from squamous (skin-type) cells; a comorbidity of some genes discussed.
Transcriptome	The complete set of RNA transcripts (gene expression profile) in a sample.
Upregulated / downregulated	Expressed at higher / lower levels in psoriatic than in normal skin.

3. Statistical Terms

Term	Definition
Accuracy	The proportion of all predictions that are correct.

Term	Definition
Adjusted (FDR-adjusted) p-value	A p-value corrected for multiple testing to control the false discovery rate.
Benjamini–Hochberg procedure	The method used to control the FDR when testing thousands of genes at once.
Bootstrap	A resampling method used here to estimate confidence intervals around the AUC.
Class imbalance	An unequal number of samples in each class (here, 92 psoriatic vs 82 normal).
Confidence interval	A range that, with stated confidence, is expected to contain the true value.
Confusion matrix	A table of true/false positives and negatives summarising classification results.
Empirical Bayes moderation	A limma technique that stabilises each gene's variance estimate using information shared across all genes.
False discovery rate (FDR)	The expected proportion of false positives among the results declared significant.
Fold change / log₂ fold change	The ratio of a gene's expression between groups; log ₂ FC places up- and down-regulation on a symmetric scale ($ \log_2\text{FC} > 1$ = at least a two-fold change).
Grid search	An exhaustive search over combinations of hyperparameter values.
Log transformation (log₂(x + 1))	A transformation that compresses skewed data onto a comparable scale.
One-standard-error rule	A rule selecting the simplest model whose error is within one standard error of the best.
Pearson correlation	A measure (−1 to +1) of the linear association between two variables.
Precision	The proportion of predicted-psoriasis samples that are truly psoriatic.
Principal Component Analysis (PCA)	A method that reduces many variables to a few axes (components) capturing the largest patterns of variation.
p-value	The probability of observing a result at least as extreme as that seen if there were no true effect.
Right skew	An asymmetric distribution with a long tail of high values, typical of RNA-sequencing data.
Sensitivity (recall)	The proportion of true psoriasis cases correctly identified.
Specificity	The proportion of true normal samples correctly identified.
Stratified split	A data split that preserves the proportion of each class in every partition.
Variance / standard deviation	Measures of how spread out a set of values is.
Volcano plot	A scatter plot of fold change against statistical significance for every gene.

4. Machine Learning and Computational Terms

Term	Definition
Alpha (α) / lambda (λ)	The elastic net's mixing parameter (L1 vs L2) and overall penalty strength.
Beeswarm plot	A SHAP plot showing every sample's per-gene contribution, coloured by expression level.
Bootstrap sample	A random sample drawn with replacement from the training data.
Classifier / classification	A model (and the task) that assigns samples to categories.
Column / row subsampling	Training each tree on a random subset of features / samples to increase diversity.

Term	Definition
Convergence	Two senses: (i) a gene supported across all three evidence lines; (ii) whether Kernel SHAP's approximation has numerically stabilised.
Cost (C) / slack variables (ξ)	The SVM trade-off between margin width and training error, and the variables permitting some misclassification.
Cross-validation	Repeatedly splitting the training data into folds to estimate performance more stably.
Data leakage	When information from the test data influences model building, giving over-optimistic performance.
Decision tree / node / leaf	A model that splits data by feature thresholds at nodes, ending in leaves that give a prediction.
Elastic net	A regularised regression combining the L1 (lasso) and L2 (ridge) penalties, used here for feature selection.
Feature	An input variable to a model (here, a gene's expression).
Feature importance	A measure of how much each feature contributes to a model's predictions.
Feature selection	Choosing a subset of informative features (genes) for modelling.
Fold	One partition of the data in cross-validation, held out in turn.
Genome-wide	Considering all expressed genes rather than a pre-defined subset.
Global / local importance	A gene's overall influence across samples / its contribution to a single prediction.
Gradient boosting / boosting rounds	Building trees sequentially, each correcting the errors of the last; boosting rounds are the number of such trees.
High-dimensional / "large p, small n"	Data with far more features (genes) than samples, prone to overfitting.
Hyperparameter	A setting fixed before training (e.g. tree depth), as opposed to parameters learned during training.
Impurity-based importance	A random forest's built-in importance measure; global and directionless.
Kernel (linear)	The function defining an SVM's decision boundary; a linear kernel gives a straight boundary.
L1 (lasso) / L2 (ridge) penalty	Penalties on the size of model coefficients; L1 can set coefficients to zero (selection), L2 shrinks them.
Learning rate (eta)	A factor shrinking each boosting tree's contribution to limit overfitting.
Min-max scaling	Rescaling features to a fixed range; more sensitive to outliers than z-score standardisation.
Naïve Bayes	A probabilistic classifier assuming features are conditionally independent (an alternative not used here).
Nested cross-validation	Cross-validation with an inner loop for model selection and an outer loop for unbiased performance estimation.
Neural network	A flexible model of connected layers (an alternative not used here).
Overfitting	When a model fits noise in the training data and generalises poorly to new data.
Permutation importance	A feature-importance method based on shuffling a feature and measuring the drop in performance.
Random forest / bagging	An ensemble of decision trees, each trained on a bootstrap sample; predictions are combined by majority vote (bagging).
Regularisation	A penalty that discourages model complexity to reduce overfitting.

Term	Definition
Shapley value / SHAP	A game-theoretic attribution assigning each feature a signed contribution to a prediction.
Standardisation (z-score)	Rescaling features to zero mean and unit variance so they are on comparable ranges.
Supervised machine learning	Learning to predict a known label (here, psoriatic vs normal) from labelled training data.
Support vector machine / margin	A classifier that separates classes with the maximum-margin hyperplane; the margin is the gap between classes.
Training / test (held-out) set	Data used to build the model / data reserved to evaluate it on unseen samples.
Tree SHAP / Kernel SHAP	Exact SHAP for tree models / model-agnostic (approximate) SHAP used here for the SVM.

5. Datasets, Databases, and Software

Name	Description
caret	R package for model training, tuning, and confusion-matrix metrics.
DISGENET	A large public database aggregating gene–disease associations, used here for annotation.
glmnet	R package implementing the elastic net for feature selection.
limma	R package for differential expression analysis using linear models and empirical Bayes moderation.
nestedcv	R package implementing the nested cross-validation framework.
pROC / mltools	R packages for ROC/AUC (with bootstrap CIs) and the Matthews correlation coefficient.
RStudio / R	The programming environment and language in which all analyses were performed.
shapviz / treeshap / kernelshap	R packages for computing and visualising SHAP explanations.
xgboost / ranger / kernlab	R packages for the XGBoost, random forest, and SVM classifiers respectively.

6. Genes Referenced in the Text

Gene functions and disease associations below are as reported in this dissertation; “novel” denotes no recorded psoriasis association in DISGENET.

Gene	Description
<i>ABCG4</i>	Candidate biomarker; upregulated; the only gene ranked in all three models’ top ten; no recorded psoriasis association (novel), strongest DISGENET links to prostate cancer and Alzheimer’s disease.
<i>AKR1B10</i>	Candidate biomarker; upregulated; strongest DISGENET link to liver cancer.
<i>AKR1B15</i>	Candidate biomarker; upregulated; no recorded psoriasis association (novel).
<i>BTC</i>	Downregulated gene encoding betacellulin, involved in skin homeostasis.
<i>CDK5R1</i>	Candidate biomarker; upregulated; no recorded psoriasis association (novel); linked to Alzheimer’s disease.
<i>COBL</i>	Candidate biomarker; downregulated; no recorded psoriasis association (novel).
<i>CXCL8</i>	Upregulated gene encoding interleukin-8 (IL-8), which drives immune-cell recruitment.

Gene	Description
<i>DEFB4A</i>	Upregulated antimicrobial gene encoding human β -defensin-2 (hBD-2); the most strongly upregulated gene in this study.
<i>FABP5</i>	Candidate biomarker; upregulated; frequently selected in prior psoriasis machine learning studies.
<i>GDA</i>	Candidate biomarker; upregulated; guanine deaminase; only nominally associated with psoriasis.
<i>HOXC10</i>	Candidate biomarker; downregulated; no recorded psoriasis association (novel); linked to gastric cancer.
<i>IL1F9 (IL36G)</i>	Candidate biomarker; well-established psoriasis inflammatory gene (recorded in DISGENET as IL36G).
<i>PI3</i>	Among the most strongly upregulated genes; encodes peptidase inhibitor 3, linked to skin-cell build-up.
<i>S100A12</i>	Candidate biomarker; upregulated S100-family gene; strongest DISGENET link to inflammatory bowel disease.
<i>S100A8 / S100A9</i>	Upregulated S100-family genes whose proteins drive inflammatory signalling; S100A8 is a candidate biomarker.
<i>SERPINB4</i>	Consistently upregulated; its protein promotes pro-inflammatory pathways and immune-cell recruitment.
<i>TAP2</i>	Candidate biomarker; upregulated; strongest psoriasis association of any candidate; also linked to type 1 diabetes and multiple sclerosis.
<i>THRSP, CCL27, KRT77</i>	The three most strongly downregulated genes in this study's differential expression analysis.
<i>TPBG</i>	Candidate biomarker; upregulated; no recorded psoriasis association (novel); linked to prostate cancer.
<i>VNN3 (VNN3P)</i>	Candidate biomarker; upregulated; the random forest's most important gene (recorded in DISGENET as VNN3P).
<i>WIF1</i>	Consistently downregulated; its protein regulates keratinocyte proliferation.

Chapter 1: Introduction

This chapter introduces the study, providing an overview and background to psoriasis and its analysis (Section 1.1) before setting out the aim and objectives (Section 1.2).

1.1 Overview and Background

Psoriasis is a chronic, inflammatory skin disease driven by immune system dysfunction and is strongly influenced by genetic factors (Armstrong et al., 2025; Man et al., 2023; Yamanaka et al., 2021). The disease is clinically characterised by inflamed, scaly plaques, or skin lesions, that often cause substantial itching and pain, reducing quality of life (Chakith et al., 2025). A recent study suggests that psoriasis affects approximately 4.4% of individuals globally, establishing it as one of the most prevalent skin disorders worldwide (Skayem et al., 2025), and its burden has continued to rise, with the number of cases globally nearly doubling between 1990 and 2021 (Wei et al., 2026).

The clinical significance of psoriasis extends well beyond dermatological symptoms. Individuals with the condition face a heightened risk of developing severe health conditions, including, but not limited to, cardiovascular disease (Gisoni et al., 2020; Puig, 2017), diabetes (Gisoni et al., 2020; Mirghani et al., 2023), psoriatic arthritis (Esti et al., 2025; Perez-Chada et al., 2025), and depression (Luna et al., 2023; Siteneski et al., 2025). These conditions can shorten life expectancy and impose a substantial burden on both patients and healthcare systems.

The pathogenesis of psoriasis is driven by an overactive immune response that mistakes the body's own cells for a threat. This triggers a cascade of inflammatory signals that act as a 'false alarm', inducing skin cells to multiply at an accelerated rate and forming the visible plaques that define the disease (Armstrong & Read, 2020; Cleveland Clinic, 2022). This inflammatory cascade creates a self-sustaining feedback loop, in which activated immune and skin cells reinforce one another and sustain chronic inflammation (Benezeder & Wolf, 2019). Fundamentally, this process involves widespread disruption of normal gene activity. Therefore, characterising specific gene dysregulations and the magnitude of these alterations is essential for understanding disease mechanisms and developing more targeted and effective treatments (Saadh et al., 2025).

Gene expression can be measured using RNA sequencing (RNA-seq), a technique that captures the expression of thousands of genes at once (Khatoon et al., 2014). By comparing skin samples from psoriatic and non-psoriatic groups, differential expression analysis of RNA-sequencing data can identify genes that are dysregulated in lesional skin. Public repositories such as the National Center for Biotechnology Information's Gene Expression Omnibus (GEO) have made a large number of RNA-sequencing datasets freely available for secondary analysis (Clough et al., 2023). This dissertation utilised one such dataset, GEO series accession GSE54456 (Elder et al., 2014), which contains gene expression measurements for 92 psoriatic and 82 non-psoriatic skin biopsies, providing expression levels for 21,510 genes. Its genome-wide coverage makes it well suited to the differential expression, classification, and interpretability analyses developed in this study.

Within psoriasis research, machine learning is increasingly applied to gene expression data, and a growing number of studies now classify psoriatic from non-psoriatic skin and identify potential biomarker genes (He et al., 2024; Le et al., 2021; Tang et al., 2025). Whereas differential expression analysis identifies the many genes that differ between psoriatic and non-psoriatic skin, machine learning models identify which of those genes they draw on to distinguish psoriatic from non-psoriatic skin, thus narrowing that long list to a compact set of candidate biomarkers worth prioritising for further investigation. A separate and larger literature applies machine learning to skin images for lesion detection or severity grading (Li et al., 2024; McMullen et al., 2025), an approach that offers little insight into the molecular drivers of the disease. What remains rare, however, is not the classification task itself, but how widely the genes are searched and how closely the resulting models are examined. Most studies utilise a pre-defined category of genes rather than letting the data select from all expressed genes, and where post-hoc interpretability techniques such as Shapley Additive Explanations (SHAP) are applied, they typically explain a single model in isolation. This dissertation addresses these gaps using the GSE54456 skin RNA-sequencing dataset. It selected candidate genes genome-wide within a leakage-controlled framework and identified those that converge across three different lines of evidence: differential expression analysis, cross-validation selection stability, and machine learning feature importance. It then characterised the wider disease associations of these genes, identifying links to a range of other conditions that may warrant further investigation. The aim and objectives of this dissertation are set out in the following section.

1.2 Aim and Objectives

The aim of this study is to identify robust candidate biomarkers of psoriasis through a genome-wide search requiring convergence across differential expression analysis, cross-validation selection stability, and machine learning feature importance, and to characterise the wider disease associations of those genes.

To achieve this aim, three measurable objectives were defined:

- To identify the genes that are most strongly differentially expressed (largest $|\log_2FC|$) between psoriatic and non-psoriatic skin, among those passing adjusted $p < 0.05$ and $|\log_2FC| > 1$.
- To perform genome-wide feature selection, without restricting to a pre-defined gene category, and to evaluate and compare the classification performance of three machine learning models from different algorithm families (XGBoost, random forest, and a support vector machine (SVM)) under leakage-controlled nested cross-validation, assessing the extent to which they agree on the genes most important to their predictions.
- To assess the robustness of candidate biomarkers of psoriasis by combining differential expression analysis, cross-validation selection stability, and machine learning feature importance, and to examine their reported disease associations using DISGENET (MedBioinformatics Solutions, 2026a).

The remainder of this dissertation is structured as follows: Chapter 2 reviews the relevant literature on psoriasis-associated genes and on the application of machine learning to gene expression data, formalises the research gap, and presents the research questions. Chapter 3 describes the dataset, pre-processing, and the analytical methodology, including the leakage-controlled nested cross-validation design. Chapter 4 presents the results for each research question. Chapter 5 discusses these results in relation to each research question, comparing the findings with the existing literature. Chapter 6 draws conclusions, reflects on the limitations of the study, and identifies directions for future work.

Chapter 2: Literature Review

Building on the background established in Chapter 1, this chapter reviews the literature on gene expression in psoriasis and its analysis by machine learning. It first outlines the search strategy used to identify relevant studies (Section 2.1), then reviews the established genes dysregulated in psoriatic skin (Section 2.2.1) and the application of machine learning to psoriasis gene expression data (Section 2.2.2). Section 2.3 draws these together to identify the methodological and interpretation gaps this study addressed, and Section 2.4 sets out the research questions that the study was structured around.

2.1 Literature Search Strategy

To identify relevant literature, comprehensive searches were performed across PubMed, Google Scholar, StarPlus, and Scopus. Search terms included combinations of the following: psoriasis, gene expression, RNA sequencing, differentially expressed genes, machine learning, XGBoost, random forest, support vector machine, elastic net, SHAP, biomarker, and other health conditions. To ensure the most recent information, research published between 2020 and 2026 was prioritised. Initially, broad searches were conducted, refining queries as key ideas and papers were identified.

2.2 Review of Relevant Literature

This review considers two branches of literature: the genes established as dysregulated in psoriatic skin through differential expression analysis, and laboratory studies of how they function (Section 2.2.1). It also looks at the machine learning approaches applied to classify psoriasis from expression data (Section 2.2.2).

2.2.1 Dysregulated Genes in Psoriatic Skin: Established Drivers and Their Broader Disease Associations

The psoriasis gene expression literature is notably consistent. Despite differences in profiling technology (for example, microarray versus RNA sequencing), patient groups, and analytical pipeline (for example, normalisation and then differential expression analysis), a recurring core of dysregulated genes is reported across studies, which lends confidence that

these differences reflect disease biology rather than technical errors. It is important to note that this core set of genes has been established primarily through differential expression studies, which apply statistical analysis to identify genes whose expression differs between psoriatic and non-psoriatic skin, together with laboratory studies of how the genes and their proteins function in terms of the disease. Neither relies on machine learning, which asks a different question and is examined in Section 2.2.2. Because dysregulation magnitudes vary with profiling technology and normalisation method, this review synthesises the direction of dysregulation and the number of independent studies in which each gene is reported, while the magnitude of differential expression for this study's dataset is reported in Chapter 4. This section synthesises findings from studies investigating several of the genes most frequently reported as dysregulated in psoriatic skin: upregulated (*S100A8*, *S100A9*, *SERPINB4*, *DEFB4A*, and *CXCL8*) or downregulated (*WIF1*, and *BTC*) before considering their broader disease associations.

Upregulated genes:

S100A8 and *S100A9* are among the genes most frequently reported as upregulated in psoriatic lesions (He et al., 2025; Zhang et al., 2025; Zhao et al., 2024a). Their encoded proteins activate inflammatory signalling in both the immune cells and keratinocytes of the skin, contributing to the self-sustaining inflammation characteristic of the disease (He et al., 2025). The *S100A9* protein has further been proposed as a candidate drug target (Silva de Melo et al., 2023). Their relevance also extends beyond the skin, with elevated levels of both proteins implicated in cardiovascular disease (Cozac et al., 2025) and the lung disease chronic obstructive pulmonary disease (COPD) (Huang et al., 2020), and *S100A9* independently linked to Alzheimer's disease (Horvath et al., 2016). *SERPINB4* follows a similar pattern, being consistently upregulated (Boonpethkaew et al., 2022; Federico et al., 2022; Xiao et al., 2024). Its encoded protein has been implicated in the activation of pro-inflammatory pathways and immune-cell recruitment to the skin, both of which help sustain psoriasis (Xiao et al., 2024). Like *S100A9*, the *SERPINB4* protein may represent a promising drug target (Zhang et al., 2023). Beyond psoriasis, elevated *SERPINB4* has been reported across a range of serious conditions, including respiratory diseases such as asthma (Kelly-Robinson et al., 2021), liver cancer (Biasiolo et al., 2012), and various squamous cell carcinomas (Izuhara et al., 2018).

The antimicrobial gene *DEFB4A* is likewise consistently reported as upregulated (Furue et al., 2020; Ma et al., 2024b; Rioux et al., 2020); its protein, human β -defensin-2 (hBD-2), is an antimicrobial peptide that recruits immune cells to the skin and promotes inflammation

(Niyonsaba et al., 2004, 2007), feeding the same inflammatory cycle as the S100 proteins. Elevated hBD-2 levels have further been observed in the inflammatory disorder ulcerative colitis (Chatterjee et al., 2025) and in oesophageal cancer (Shi et al., 2014). *CXCL8*, encoding the protein interleukin-8 (IL-8), is also consistently upregulated (Boonpethkaew et al., 2022; Yang et al., 2022; Zhang et al., 2024); unlike β -defensin-2, IL-8 lacks direct antimicrobial activity but drives immune-cell recruitment, sustaining the inflammatory aspect of the disease (Liu et al., 2022). Elevated IL-8 levels have been reported across a broad range of conditions, spanning cancers such as osteosarcoma (Tatsuno et al., 2024) and papillary thyroid carcinoma (Liu et al., 2025), psychiatric and neurological conditions such as post-traumatic stress disorder (PTSD) and Alzheimer's disease (Liao et al., 2026), and inflammatory diseases such as rheumatoid arthritis (Qian et al., 2025). Taken together, these five upregulated genes converge on a common axis of amplified inflammatory signalling and immune-cell recruitment, consistent with the accelerated keratinocyte proliferation and plaque formation that define the disease (Benezeder & Wolf, 2019).

Downregulated genes:

The downregulated genes tell a complementary story of lost regulatory control across two processes: keratinocyte proliferation and epidermal homeostasis. *WIFI* is consistently reported as downregulated in psoriatic lesions (Gudjonsson et al., 2010b; Ma et al., 2024a; Zhang et al., 2025); its protein is thought to regulate keratinocyte proliferation, such that its reduced levels may contribute to the epidermal hyperproliferation underlying plaques (Liu et al., 2023), and its downregulation is also implicated in several cancers, including cervical (Liang et al., 2016), gastric (Chai et al., 2025), and pancreatic malignancies (Zhu et al., 2017). *BTC*, encoding betacellulin, is likewise frequently downregulated (Gudjonsson et al., 2010a; Peng et al., 2022; Suárez-Fariñas et al., 2012). Betacellulin participates in skin homeostasis, and its reduced expression is associated with increased plaque severity (Peng et al., 2022). While fewer studies have examined its broader disease associations compared to the other proteins discussed, reduced levels of betacellulin have been linked to schizophrenia (Cota-Coronado et al., 2025). Whereas the upregulated genes reflect active inflammation, these downregulated genes reflect a weakening of the normal constraints on keratinocyte proliferation and epidermal homeostasis. Across both groups, the consistent implication of these genes in conditions beyond the skin provides molecular support for the systemic comorbidities associated with psoriasis, and it is this cross-disease pattern that motivates the

present study's later cross-referencing of its own candidate genes against the DISGENET gene–disease association database (MedBioinformatics Solutions, 2026a).

However, the very consistency that lends this literature its strength also exposes its principal limitation. These findings rest mostly on differential expression analysis. Differential expression analysis reliably identifies genes that are consistently dysregulated in psoriasis, and this study utilised it for exactly that purpose. What it does not establish, however, is predictive importance: that a gene's expression differs between lesional and healthy skin is a statistical association, not evidence that a machine learning model will rely on it to classify. A regularised model such as the elastic net selects genes for their contribution to prediction, retaining correlated groups of informative genes together rather than ranking them by degree of dysregulation (Zou & Hastie, 2005). A highly dysregulated gene may therefore carry little additional predictive weight once genes that are correlated with it are already in the model, and studies find that the genes most useful for classification are not always the most statistically differential (Das et al., 2020). Identifying which genes a model relies on to classify psoriasis, rather than ones that are merely dysregulated, is therefore a distinct and complementary task. This task prioritises a compact set of candidate biomarkers that differential expression analysis alone would not single out. Such a shortlist is better suited to the laboratory studies that could test whether these genes contribute to psoriasis, a question neither differential expression analysis nor classification can answer on their own. It is this predictive task that the following section examines.

2.2.2 Machine Learning Applied to Psoriasis Gene Expression

Where differential expression analysis asks which genes differ in expression between psoriatic and non-psoriatic skin, supervised machine learning asks a distinct question: which genes a model relies upon to separate the two groups. Over the past five years, this framing has been applied with growing frequency to psoriasis gene expression studies, with a substantial recent literature training classifiers to distinguish psoriatic from non-psoriatic skin (Deng et al., 2024; He et al., 2024; Le et al., 2021; Liu et al., 2026; Ma et al., 2024b; Tang et al., 2025). Classifying psoriatic from healthy skin is an easy task for supervised machine learning models when training and testing sets draw from the same cohort, with reported performance at or near the ceiling. Le et al. (2021) trained four classifiers (k-nearest neighbours, naïve Bayes, random forest, and a support vector machine) on a merged microarray cohort of 286 samples (GDS4600 and GDS4602 combined), with random forest reaching 98.3% accuracy, 98.6% recall, and 98% precision in repeated five-fold cross-validation. He et al. (2024) fitted nine models to a six-

gene panel derived from CD8⁺ T cells and evaluated them on a held-out 30% split of the same dataset used for this study, GSE54456, where random forest and the support vector machine returned 1.000 on essentially every performance metric. Tang et al. (2025) trained five models on 34 housekeeping genes using an 80/20 split of the GSE54456 cohort, and all five reached an area under the curve and F1 of 1.0 with 100% accuracy on the internal split. This separability arises because psoriatic and non-psoriatic skin differ substantially in their gene expression, with thousands of genes significantly differentially expressed, and these fall into several distinct modules of co-expressed genes (Li et al., 2014). Because genes within each module are co-expressed, much of the classification signal is shared among correlated genes rather than concentrated in a few, a well-documented property of high-dimensional expression data in which correlated genes carry overlapping predictive information (Ding & Peng, 2003). Consequently, many different gene sets, including small panels, separate the classes perfectly or near-perfectly within a cohort. Le et al.'s (2021) random forest model obtained almost perfect cross-validated performance from 54,000 probes, their 35-gene panel, and a three-gene panel (AUC 0.979, 0.998, and 0.977 respectively). He et al. (2024) achieved comparably strong performance across nine models utilising only a six-gene panel.

Whether these gene panels carry predictive signal that transfers or merely separates the two groups well in one particular dataset, only becomes clear when the models are tested on external data. Tang et al. (2025) validated on six independent datasets and found that the models, indistinguishable on the internal split, diverged sharply. The averaged neural network (avNNet) held above 90% across the external skin datasets, whereas the support vector machine fell below 75% on some sets. Not every study finds such a large gap. Le et al. (2021) validated on two independent microarray sets, where random forest reached 96.7% accuracy, 93.9% recall, and 100% precision on GDS3539 and 100% across all these metrics on the smaller GDS4891, while He et al. (2024) performed no external validation at all. However, the possibility that internally strong models can differ sharply in how well they generalise is precisely why within-cohort performance is weak evidence about which genes are predictively important. Strong within-cohort performance is nearly guaranteed by the strength of the classification signal, and only external validation distinguishes models that generalise from those that do not. The informative question is therefore not whether psoriasis can be classified, but which genes a study surfaces, and how robustly they hold up.

Furthermore, the genes a study reports depend heavily on which genes it allowed the models to consider. Most psoriasis machine learning studies do not search genome-wide. They restrict the candidate pool to a pre-defined biological category before any classifier is trained,

such as housekeeping genes (Tang et al., 2025), lysosomal genes (Deng et al., 2024), apoptosis genes (Liu et al., 2026), lactylation genes (Li et al., 2026), neutrophil genes (Zhao et al., 2024b), or genes tied to a specific immune cell type such as CD8⁺ T cells (He et al., 2024) or dendritic cells (Ma et al., 2024b). This pool is then narrowed further by a combination of differential expression analysis, network analysis, and feature selection algorithms such as LASSO, random forest, and SVM-RFE. Because correlated genes carry overlapping signal, this pre-restriction does not harm classification performance metrics on models that were trained and tested on the same cohort. They still perform well, but it largely determines which genes are found, since a model can only identify genes that it was given. Reported biomarkers are therefore shaped in part by the starting pool, a limitation examined further in Section 2.3.

Additionally, where a study does ask which genes a trained model relied upon, it typically interprets a single model. Under correlated features, importance rankings from a single model are biased and can mislead interpretation, because correlation between features distorts the importance each is assigned (Tološi & Lengauer, 2011). He et al. (2024) trained nine models but applied Shapley Additive Explanations (SHAP) to only one, and Liu et al. (2026) applied Descriptive Machine Learning Explanations (DALEX) to a single gradient-boosted model, while other studies report predictive performance with no post-hoc interpretation at all (Deng et al., 2024; Tang et al., 2025). Whether feature importance rankings agree across independent models is rarely tested, even though different algorithms are known to select different features as important (Saarela & Jauhiainen, 2021). A gene ranked highly across several model types is therefore less likely to reflect the bias of any one algorithm and is a stronger candidate. In addition, drawing on several models rather than one also reduces the risk of missing informative genes, since a gene overlooked by one algorithm may be ranked highly by another.

Against this backdrop of overlapping signal across correlated genes and pool-dependent selection, it is notable that specific genes recur across independent pipelines despite the many alternatives available. One of the most consistent is *FABP5*, retained as a final selected biomarker across different starting gene pools and independent feature selection pipelines, including a genome-wide random forest selection (Le et al., 2021), a dendritic-cell focused combined LASSO, random forest, and SVM-RFE pipeline (Ma et al., 2024b), and a random forest and LASSO feature selection intersection applied to lactylation related genes (Li et al., 2026). Two further genes surface at the feature selection stage without always being retained, and their behaviour illustrates how selection depends on the algorithm and dataset used. In a neutrophil-gene dataset, *SI00A12* was selected by LASSO but did not survive the subsequent

SVM-RFE step (Zhao et al., 2024b). In a separate study, both *AKR1B10* (under LASSO and SVM-RFE) and *SI00A12* (under SVM-RFE) appeared among the selected features, yet neither was retained among the final markers (Hao et al., 2024). Their appearance under some selectors but not others, within a single study, is itself a small illustration of how correlated, overlapping signal makes selection algorithm dependent. That *FABP5* nonetheless recurs across studies, despite each drawing on a different gene pool, suggests it is a strong candidate biomarker. Whether recurrence holds under a genome-wide search, with agreement across independent models on which genes are most important, is a question this literature does not resolve, and one the present study is designed to address.

2.3 Research Gap

As Section 2.2.2 highlighted, the strength of the classification signal and its spread across many correlated genes mean that high within-cohort performance metrics are expected, so the limitations of the previous studies lie not in the performance metrics but in two connected areas: a methodological gap in how genes are selected for analysis, and an interpretation gap in how the resulting models are examined. This study addressed both, searching genome-wide rather than within a pre-restricted category, and assessing agreement across different algorithm families on which genes are most important, rather than interpreting a single model.

A methodological gap in feature selection: As Section 2.2.2 established, most existing psoriasis machine learning studies restrict the candidate space before modelling, first to a pre-defined biological category such as lysosomal (Deng et al., 2024), apoptosis (Liu et al., 2026) or CD8⁺ T-cell genes (He et al., 2024), then narrowing further through a combination of differential expression analysis, network analysis, and feature selection. However, this limits what can be discovered. A model can only identify genes fed into it beforehand, so any informative genes outside the chosen category were excluded by construction. This study instead applied elastic net feature selection across 17,656 genes, letting predictive value alone select a compact panel without restricting to a pre-defined category, allowing candidates outside conventional categories to emerge. The panel was then used to train the classification models.

An interpretation gap in model explainability: Where post-hoc explainability is applied at all, it is applied to a single model over an already restricted candidate space. He et al. (2024) trained nine models but applied SHAP to only one of them, and Liu et al. (2026) applied Descriptive Machine Learning Explanations (DALEX) to a single gradient-boosted

model among the several models that they trained. Other studies reported predictive performance with no post-hoc explainability step whatsoever (Deng et al., 2024; Tang et al., 2025). Whether independent algorithms agree on which genes matter is therefore rarely tested, even though different algorithms can select different features as important (Saarela & Jauhiainen, 2021). Because correlation between features distorts the importance a single model assigns (Tološi & Lengauer, 2011), a gene ranked highly across several model types is less likely to reflect the bias of any one algorithm and is therefore a stronger candidate. Drawing on several models rather than one also reduces the risk of missing informative genes, since a gene overlooked by one algorithm may be ranked highly by another. The study described in this dissertation undertook a comparison of SHAP-based feature importance across three different classification algorithm families, XGBoost (boosting), random forest (bagging), and a support vector machine (margin-based). They were trained under identical conditions, so that the extent to which the models agreed on the most important genes could be assessed.

These limitations define the gap that this study addressed; prior works have classified psoriasis from gene expression data, but from pre-selected genes, usually from a single biological category, and post-hoc model interpretation was confined to either a single model or was absent altogether. This study utilised GEO's publicly available GSE54456 RNA-sequencing dataset (Elder et al., 2014), comprising 92 psoriatic and 82 non-psoriatic skin biopsies, to identify candidate biomarkers that converge across differential expression analysis, cross-validation selection stability, and machine learning feature importance.

2.4 Research Questions

Following the identification of gaps in the literature, this study was developed to answer the following research questions:

RQ1: Which genes show the largest differential expression between psoriatic and non-psoriatic skin, among those passing an adjusted $p < 0.05$ and $|\log_2FC| > 1$ threshold?

RQ2: How accurately can machine learning models classify psoriasis from gene expression data under leakage-controlled nested cross-validation, and to what extent do different algorithm families agree on the same genes as most important to their predictions?

RQ3: Which genes are supported as robust candidate biomarkers of psoriasis through convergence of differential expression analysis, cross-validation selection stability, and machine learning feature importance, and what disease associations do they have?

The methodology designed to address these questions is set out in the following chapter.

Chapter 3: Methodology

Having reviewed the literature in Chapter 2, and identified research questions, this chapter describes the data and analytical methodology used to address the research questions. It first outlines the methodological approach and dataset (Section 3.1), initial data quality checks (Section 3.2), and exploratory data analysis (Section 3.3). Sections 3.4 to 3.6 then set out the three analyses aligned with the research questions: differential expression analysis (Section 3.4, RQ1), the machine learning framework (Section 3.5, RQ2), and the identification of candidate biomarkers (Section 3.6, RQ3). Section 3.7 details the software and implementation, and Section 3.8 addresses ethical considerations.

3.1 Methodological Approach and Dataset Overview

This study followed a deductive approach, utilising established research on the genes associated with psoriasis to inform the analytical framework (Saunders et al., 2016). It employed a quantitative methodology, drawing inferences from the statistical and computational analysis of numerical gene expression measurements rather than from qualitative interpretation. This dissertation is a secondary analysis of the RNA sequencing dataset GSE54456, publicly available from the NCBI Gene Expression Omnibus website (GEO) (Elder et al., 2014). The dataset contains gene expression data for 21,510 genes across 174 skin samples (92 psoriatic and 82 normal). Expression values are given as Reads Per Kilobase per Million mapped reads (RPKM), a measure of gene expression activity derived from RNA sequencing (Zhao et al., 2020).

The file *RPKM Values.txt* was downloaded directly from the GEO website for GSE54456 (Elder et al., 2014) and contained the RPKM values for all 21,510 genes across the 174 skin samples, with each column identified by its sample ID. The corresponding tissue type ('Psoriatic' or 'Normal') was not included in this file and was compiled separately from the metadata on GSE54456's GEO page into a CSV file called *Sample ID and Tissue Type CSV File*. These two files are on the Google Drive. The two files were then merged in RStudio on Sample ID, producing a single analysis table of 21,510 genes across 174 samples, each labelled 'Normal' or 'Psoriasis'. All analyses were conducted in RStudio (version 2025.09.0+387). The

dependent variable was tissue type. The independent variables were the genes. The class distribution was mildly imbalanced but close to being evenly distributed with 82 ‘Normal’ (47.1%) and 92 ‘Psoriasis’ (52.9%) samples. Table 3.1 summarises the structure of the dataset.

Table 3.1: *Structure of the merged dataset (174 samples).*

Variable	Role	Data type	Values	Missing (%)
Genes	Independent Variables	Continuous (Numeric)	RPKM values for 21,510 genes	0%
Tissue type	Dependent Variable	Binary Categorical	Normal Psoriasis	0%
Sample ID	Identification	Nominal Categorical	174 unique values (For example, ‘CASP1234’ or ‘M3059’)	0%

3.2 Initial Data Quality

Before any analysis, a series of integrity checks were run to confirm that the dataset was fit for modelling (Table 3.2). No missing values were present for Sample ID, Tissue Type, or any of the gene variables, so no imputation was required. Genes with an RPKM value of zero were retained as valid measurements, since in RPKM data, a zero denotes a gene not detected in that sample rather than a missing value. All 174 samples had unique Sample IDs, ruling out the risk of the same sample appearing in both the training and testing partition. No sample had zero expression across all genes, confirming there were no empty samples. However, 411 genes showed zero expression across all samples. These were uninformative and were removed. The cleaned dataset was therefore comprised of 174 samples and 21,099 genes.

Table 3.2: *Initial data quality checks and the resulting cleaning decisions.*

Quality check	Result	Action taken
Missing values (for Sample ID, Tissue Type, or any of the genes)	0	None required
Duplicate Sample IDs	0	None required
Genes with zero expression in all samples	411	Removed (uninformative)
Samples with zero expression for all genes	0	None required
Genes retained for analysis	21,099	Carried forward

3.3 Exploratory Data Analysis

Exploratory data analysis was performed to examine the distribution of the gene expression values, confirm data quality, identify anomalous samples, and assess whether tissue type was the dominant source of variation in gene expression across the samples before modelling. Five analyses were conducted: distributional summary statistics (Section 3.3.1), density plots (Section 3.3.2), per-sample box plots (Section 3.3.3), sample-to-sample correlation (Section 3.3.4), and Principal Component Analysis (PCA) (Section 3.3.5).

3.3.1 Descriptive Statistics

Descriptive statistics were computed across all 21,099 genes and 174 samples (Table 3.3). The mean gene expression level was 23 RPKM, almost eight times the median of 3 RPKM, and the standard deviation (238 RPKM) and maximum value (35,651 RPKM) were extremely large relative to the median. A mean expression that greatly exceeds the median indicates a strong positive right skew in the distribution of the gene expression values. The dataset also contained numerous high outliers (maximum was 35,651 RPKM). They were all retained rather than removed, as they represented genuine high-expression genes rather than technical errors. This was consistent with the original study of this dataset (Li et al., 2014), which reported all of the RPKM values and used them for analysis without removing high-expression outliers.

Table 3.3: *Descriptive statistics of gene expression (RPKM) computed across all 21,099 genes and 174 samples.*

Statistic	Expression Level (RPKM)
Mean	23
Median	3
Standard Deviation	238
Minimum	0
Maximum	35,651

3.3.2 Density Distribution and the Rationale for Logarithmic Transformation

The overall distribution of gene expression values was visualised using kernel density plots for each tissue type (Figure 3.1). Expression density was concentrated near zero with an extremely long right tail extending to over 35,000 RPKM, confirming the strong right skew implied by the descriptive statistics (Section 3.3.1). This shape is characteristic of RNA-sequencing data, in which expression spans a large, dynamic range, with most genes expressed at low levels, while a small proportion of highly expressed genes produce very large values that dominate the upper tail and skew the overall distribution (Conesa et al., 2016). A $\log_2(x + 1)$ transformation was therefore applied. It was necessary for limma because, in the untransformed data, genes with higher average expression tend to have higher variance, which violates the constant variance assumption that limma relies on (Law et al., 2014). The transformation was also applied for the elastic-net feature selection, which is distorted by the skewness and extreme values of untransformed RNA-sequencing data and tends to preferentially select high-variance genes (Zwiener et al., 2014). It was likewise appropriate for the SVM classification model which, as a margin-based method, can be influenced by extreme values. No transformation was applied to the tree-based classification models (random forest and XGBoost), which split on the ordering of feature values rather than their magnitude, and are therefore unaffected by monotonic transformations such as logarithmic ones (Pinheiro et al., 2025).

After the $\log_2(x + 1)$ transformation, the extreme right-skew of the raw RPKM values was substantially reduced and the data compressed onto a comparable scale (Figure 3.2). The ‘Normal’ and ‘Psoriasis’ densities were almost identical, indicating that psoriasis does not shift the overall gene expression distribution. Instead, disease-related differences occur at the level of specific genes, consistent with the large scale, gene specific dysregulation reported for psoriasis (Li et al., 2014).

Figure 3.1: Kernel density distribution of untransformed gene expression values (RPKM) by tissue type (x-axis truncated at 500 for readability; values extend to 35,651 RPKM). The extreme right skew motivated the $\log_2(x + 1)$ transformation.

Untransformed RPKM Gene Expression Density Distribution

Extreme right skew confirms the need for $\log_2(x+1)$ transformation.
X-axis limited to 0-500 for readability — values extend to 35,651.

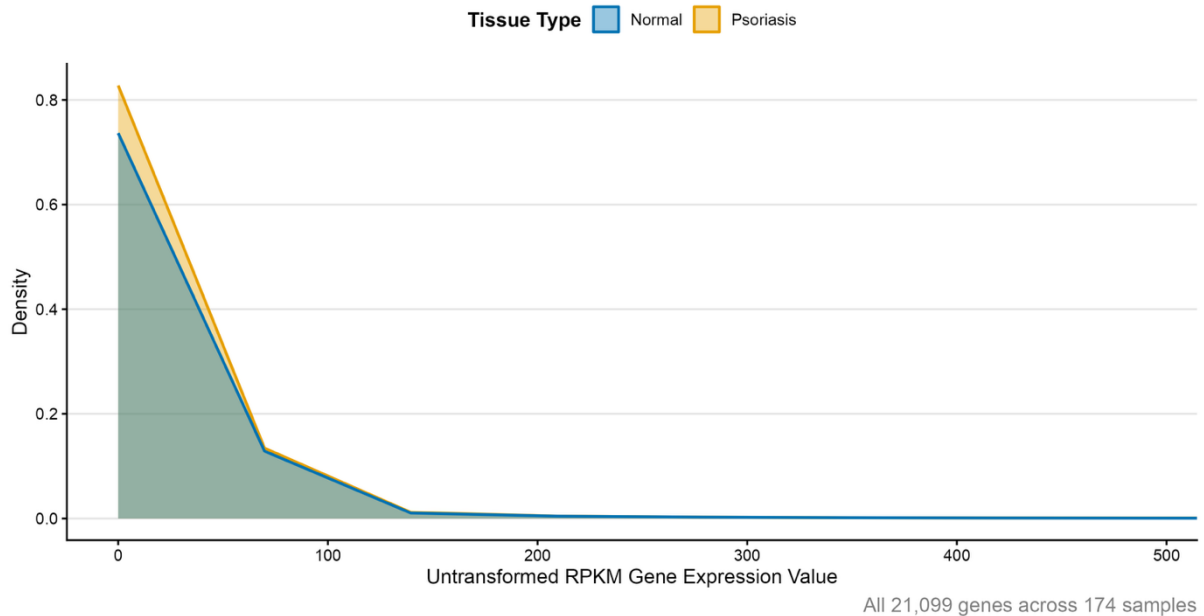
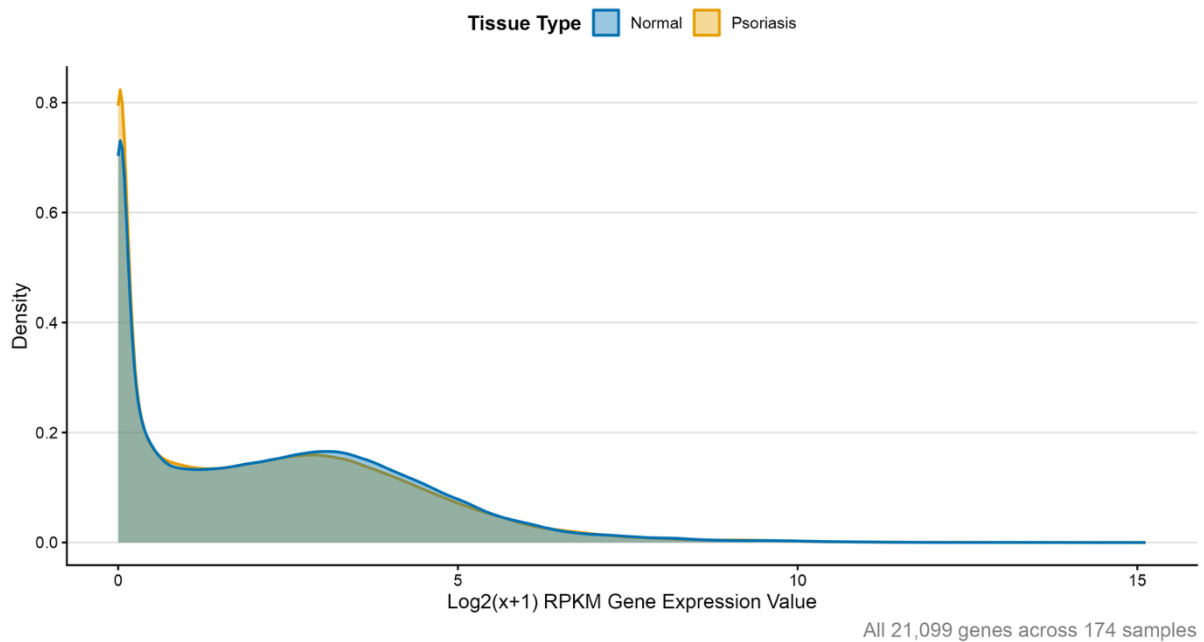


Figure 3.2: The transformation greatly reduced the right skew seen in Figure 3.1. The resulting distribution was bimodal, reflecting a large group of low-expression genes and a second group of more highly expressed genes.

$\log_2(x+1)$ Transformed RPKM Gene Expression Density Distribution

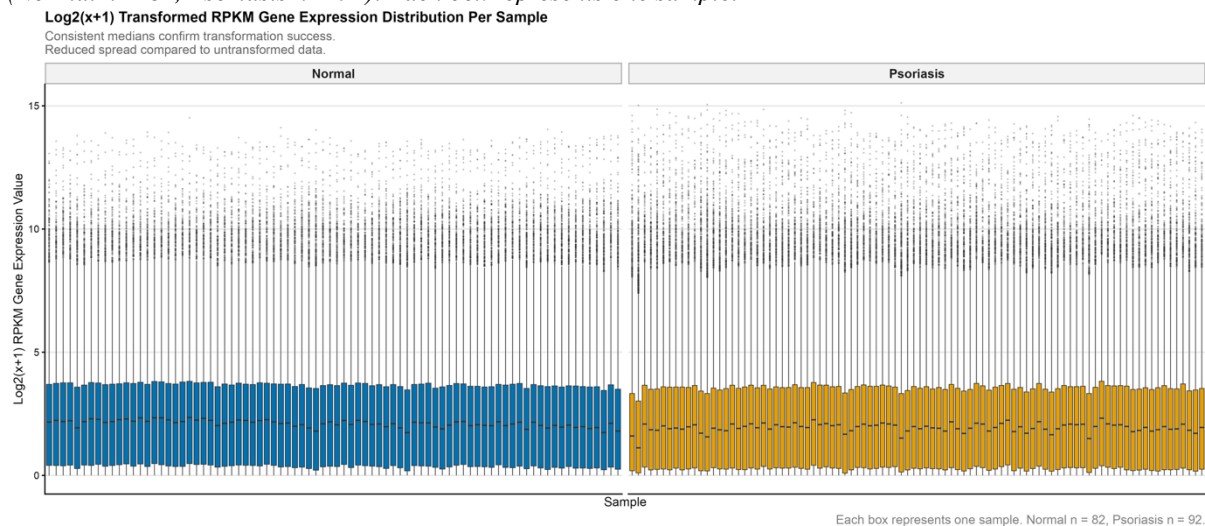
Extreme right skew of untransformed RPKM data successfully reduced by $\log_2(x+1)$ transformation



3.3.3 Per-sample Distribution

Quality control is a standard step in RNA sequencing analysis (Conesa et al., 2016). Per-sample box plots were therefore produced (Figure 3.3) to screen for anomalous samples, such as those affected by technical issues, which would appear as a shifted median or unusual spread relative to the rest of the cohort. No such sample was evident. Medians and interquartile ranges were broadly similar across all 174 samples in both tissue groups, with no sample deviating markedly from the rest, indicating similar overall gene expression distributions and providing no reason to exclude any samples. The points above each box are highly expressed genes lying beyond the whiskers, an expected feature of gene expression data rather than a sign of error (Conesa et al., 2016).

Figure 3.3: Per-sample distribution of $\log_2(x + 1)$ transformed gene expression (RPKM) by tissue type (Normal $n = 82$, Psoriasis $n = 92$). Each box represents one sample.



3.3.4 Sample-to-Sample Correlation

As a further check, the correlation between samples was examined to confirm that each sample's overall gene expression profile resembled the cohort. This complemented what was found by the box plots (Section 3.3.3). The Pearson correlation was calculated between every pair of samples, and each sample was summarised by its mean correlation with the other 173. A sample with a technical error would correlate poorly with the rest and stand out with a low mean correlation value. All samples correlated highly (range: 0.89 – 0.97, cohort mean: 0.96). The lowest five mean correlations are shown in Table 3.4 below. The lowest belonged to sample M3171 (mean correlation = 0.89), which sat slightly below the others, but its mean correlation remained high, far from the low values that would indicate a technical failure. It was retained and is examined further in the PCA below.

Table 3.4: *The five samples with the lowest mean pairwise correlation to the rest of the cohort (174 samples; cohort mean = 0.96).*

Sample ID	Tissue type	Mean correlation
M3171	Psoriasis	0.89
M4294	Psoriasis	0.93
M4310	Psoriasis	0.93
M4505	Psoriasis	0.94
M4344	Psoriasis	0.94

3.3.5 Principal Component Analysis

Principal Component Analysis (PCA) reduces the thousands of genes to a few axes that capture the largest patterns of variation in gene expression among samples, allowing the main structure of the data to be visualised and any sample with an unusual overall gene expression profile to be identified (Jolliffe & Cadima, 2016). The first principal component (PC1) accounted for 41.4% of the total variance in gene expression and clearly separated psoriatic from normal skin samples (Figure 3.4). That a single component accounts for such a large share of the variation and aligns with tissue type confirms that the disease is the dominant source of variation in gene expression, and that the two groups are transcriptionally distinct. This anticipated the strong classification performance reported later in this dissertation. While a small number of samples were on the edges of their group's confidence region, M3171 was the most extreme. It lay well outside the psoriasis ellipse on PC2 and was flagged by the correlation check (Section 3.3.4). A genuine outlier should be retained rather than deleted, as removing accurate observations biases results (Aguinis et al., 2013). M3171 showed no sign of technical failure. It separated correctly along the disease axis (PC1), and its mean correlation with the other samples was only marginally below the cohort average (mean correlation of 0.89 versus cohort mean correlation of 0.96, Section 3.3.4). M3171 was therefore retained.

Figure 3.4: PCA of $\log_2(x + 1)$ RPKM gene expression ($n = 174$; 82 Normal, 92 Psoriasis). PC1 (41.4% of variance) separates by tissue type; PC2 (14.2% of variance) is shown on the vertical axis; dashed ellipses are 95% confidence regions. M3171 (labelled) is an atypical but retained psoriasis sample.

PCA Graph of Log2(x+1) RPKM Gene Expression Across All Samples

Separation along PC1 indicates the disease signal dominates the variation.



Each point is one sample ($n = 174$; 82 Normal, 92 Psoriasis) across 21,099 genes. Dashed ellipses: 95% confidence region per group. M3171 is an atypical but retained psoriasis sample.

3.4 Differential Expression Analysis (RQ1)

To address RQ1, differential expression between psoriatic and normal skin was assessed across all 21,099 retained genes and 174 samples using limma, a widely used method for differential expression analysis of gene expression data (Ritchie et al., 2015). For each gene, limma fits a linear model testing whether expression differs between the two tissue types, with ‘normal’ as the reference group. Limma stabilises each gene’s variance estimate using the overall variance pattern across all genes, an approach called empirical Bayes moderation, so that variance estimates for individual genes are more stable than those from a standard per-gene t-test (Smyth, 2004). This is particularly valuable when per-gene variance must be estimated from a limited number of samples.

Differential expression analysis from RNA sequencing is traditionally performed on raw read counts, using software that models them with the negative binomial distribution (Ritchie et al., 2015), such as the R packages DESeq2 and edgeR. The data here, however, were not raw counts but RPKM values (Li et al., 2014), continuous, normalised measures rather than

the integer read counts those count-based tools require. Limma's linear modelling of log-transformed expression was therefore more appropriate.

Testing 21,099 genes at once means that some would appear statistically significant purely by chance. To account for this, p -values were adjusted using the Benjamini-Hochberg procedure, which controls the false discovery rate, defined as the expected proportion of false positives among the genes flagged as being statistically significant (Benjamini & Hochberg, 1995). Two criteria were then combined to define differential expression. First, a false discovery rate (FDR)-adjusted p -value below 0.05 indicated that a gene's difference in expression between psoriatic and normal skin was statistically significant at a 5% false discovery rate. Second, an absolute $\log_2$ fold change above 1 required that difference to be of a meaningful magnitude, capturing genes at least twice as highly expressed (upregulated) or at least twice as lowly expressed (downregulated) in psoriatic versus normal skin. Requiring both guards against differences that are statistically detectable but biologically trivial. These criteria (FDR-adjusted $p < 0.05$ and $|\log_2\text{FC}| > 1$) match those applied by Deng et al. (2024) on comparable psoriasis expression data, and the original analysis of this dataset (Li et al., 2014) also used the $|\log_2\text{FC}| > 1$ threshold. Results were visualised as a volcano plot and exported as ranked gene lists, with the top 10 upregulated and downregulated genes by fold change reported in Chapter 4.

3.5 Machine Learning Framework (RQ2)

This section sets out the machine learning framework used to address RQ2. It describes the stratified 80/20 data partitioning (Section 3.5.1) and low- and zero-variance feature filtering (Section 3.5.2), before introducing the three classification models compared (Section 3.5.3). The leakage-controlled nested cross-validation design is then detailed (Section 3.5.4), including the elastic net feature selection (Section 3.5.4.1) and feature standardisation and hyperparameter tuning (Section 3.5.4.2) performed within each training fold, followed by the performance metrics (Section 3.5.5) and SHAP-based feature importance (Section 3.5.6).

3.5.1 Data Partitioning (80/20 Stratified Split)

Before any model development, the 174 samples were divided into a training set and a held-out test set using an 80/20 split. This allocated 140 samples (66 normal, 74 psoriatic) for training and 34 (16 normal, 18 psoriatic) for testing. Separating the data before any further processing is essential to obtain an unbiased estimate of how the models generalise to unseen samples: if information from the test set influences pre-processing, feature selection, or model

tuning, performance estimates become optimistically biased through data leakage (Pinheiro et al., 2025). The test set was therefore set aside immediately after the split and used only once, for the final evaluation. The split was stratified by tissue type so that the proportion of psoriatic and normal samples in the training and test sets matched that of the full dataset (roughly 53% psoriatic and 47% normal). This prevents the distorted class ratios that a purely random split can produce by chance (Raschka, 2018), which would give unrepresentative performance estimates. The 80/20 ratio is a widely used convention (Raschka, 2018).

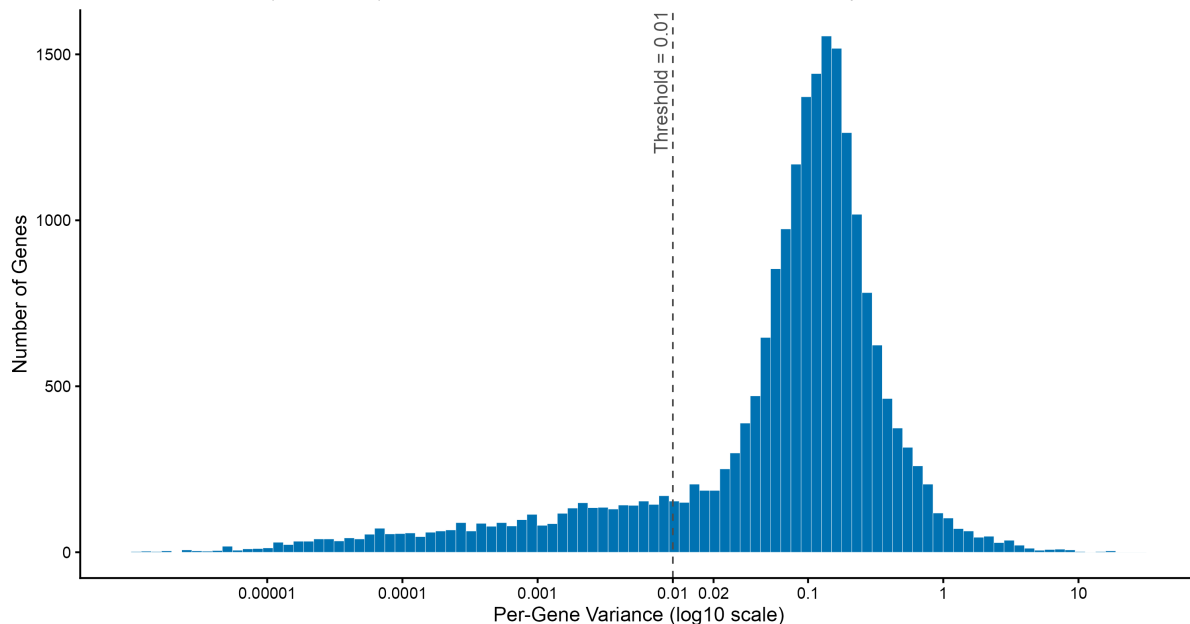
3.5.2 Low and Zero Variance Feature Filtering

Following the split, genes with little or no variation in gene expression across the training samples were removed. A near constant or constant gene cannot help a classifier distinguish psoriatic from normal tissue but still adds noise and computational cost. This is a particular concern here, where the genes vastly outnumber the samples. Removing such uninformative features is a standard step in high-dimensional data that reduces overfitting (Bommert et al., 2021). Per-gene variance was computed on the $\log_2(x + 1)$ training data only, and genes with a variance at or below 0.01 were removed. Computing the filter from the training samples alone avoids data leakage: deriving it from the full dataset would let information about the held-out test samples influence which genes go into the models (Pinheiro et al., 2025). The same filter was then applied to both the training and test sets, so the test set was filtered using only information from the training data. The 0.01 threshold was chosen from the distribution of per-gene variances (Figure 3.5), which shows the low or zero-variance shoulder of genes separated from the main peak of informative genes, with the threshold falling in the trough between them. The cut-off therefore reflected the structure of the data rather than an arbitrary choice. The filter removed 3,443 genes with low or zero-variance, leaving 17,656 for model development.

Figure 3.5: Per-gene variance in expression across the 140 training samples. Computed on $\log_2(x + 1)$ RPKM expression, shown on a $\log_{10}$ x-axis for readability, as the variances span several orders of magnitude. The dashed line marks the 0.01 filtering threshold. Genes with zero variance cannot be shown on a log axis and are not plotted but they were removed by the filter.

Distribution of Per-Gene Variance (Training Set)

A variance threshold (dashed line) removes the low-variance shoulder; the informative peak is retained.



3.5.3 Classification Models

Three algorithms, a random forest, XGBoost, and a linear support vector machine were selected because they classify using different mechanisms (bagging, boosting, and margin maximisation). Each is defined and compared in Table 3.5. The alternative methods that were considered are listed in Table 3.6.

Table 3.5: Comparison of the three classification models. Each shaded band gives the model, its formula and the definition of every term; the three columns beneath compare the model's suitability for this dataset, prior use on psoriasis gene expression data, and its limitations.

Suitability for this dataset	Prior use on psoriasis gene expression data	Limitations
Random forest: A random forest aggregates B decision trees ($T_1 \dots T_B$), each grown on a bootstrap sample of the training data with only a random subset of features considered at each split; the class prediction is the majority vote across all trees (Breiman, 2001): $\hat{y} = \text{mode}\{T_1(x), \dots, T_B(x)\}$ where x is the gene expression profile of a sample, $T_B(x)$ is the class predicted for x by the b -th tree, and $\hat{y}$ is the forest's final predicted class (the most frequently predicted class across the B trees).		
It makes no assumptions about how the data are distributed and captures non-linear, interacting relationships	Has been applied successfully to psoriasis gene expression classification in past studies, including on the same dataset	Built-in impurity feature importance is global and directionless. It indicates which genes influence

Suitability for this dataset	Prior use on psoriasis gene expression data	Limitations
between variables (Krennmair & Schmid, 2022). This is relevant for this dataset, as the expression data are non-normally distributed (Figure 3.1, Section 3.3.2) and genes relate to disease status through complex, non-linear effects rather than simple independent ones (Li et al., 2014). Secondly, because tree splits depend only on the ordering of predictor values rather than their magnitude, the random forest is invariant to monotonic per-feature transformations and so requires neither standardisation nor log transformation (Pinheiro et al., 2025). This suits the data, whose expression spans several orders of magnitude (Figure 3.1, Section 3.3.2). It also avoids fitting a data-estimated pre-processing step (standardisation), removing one potential source of data leakage.	used in this dissertation (GSE54456), where He et al. (2024) reported perfect within-cohort classification, consistent with the uniformly high within-cohort performance expected on psoriasis gene expression data (Section 2.2.2). Le et al. (2021), using a different dataset, similarly found random forest to be their strongest classifier.	predictions overall, but not whether a gene drives predictions towards psoriasis or normal. Addressed via post-hoc SHAP (Section 3.5.6).
XGBoost: XGBoost is a gradient-boosted tree ensemble that builds trees additively, each new tree fitting the residual error of the current model (Chen & Guestrin, 2016): $\hat{y} = \sum_k f_k(x), \text{ minimising } \mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \text{ where } \Omega(f) = \gamma T + \frac{1}{2} \lambda \ w\ ^2$ Here f_k is the k-th of K trees in the ensemble, ℓ is the loss function measuring the difference between the true label y_i and the prediction $\hat{y}_i$, and Ω penalises model complexity through the number of leaves T (weighted by γ) and the magnitude of the leaf weights w (weighted by λ).		
- It makes no assumptions about how the data are distributed and captures non-linear, interacting effects between genes (Hastie et al., 2009). - Tree-based methods' splits respond to the ordering of predictor values rather than their magnitude (Pinheiro et al., 2025), so no standardisation or log transformation was needed, thus avoiding a pre-processing	Has been applied successfully to psoriasis classification in past studies, including on this dataset: He et al. (2024) reported strong performance across all metrics (approximately 0.95–1.00)	More tuneable hyperparameters than the other models, raising overfitting risk at this sample size. Tuning was limited to the most influential, with overfitting controlled by favouring simpler models, shallow trees, few boosting rounds, column subsampling, and a low learning rate.

Suitability for this dataset	Prior use on psoriasis gene expression data	Limitations
step (standardisation) where data leakage can arise.		
Linear SVM: A linear SVM finds the hyperplane that separates the two classes with the maximum margin (Cortes & Vapnik, 1995): $\text{minimise } \frac{1}{2}\ w\ ^2 + C \sum_i \xi_i \text{ subject to } y_i(w \cdot x_i + b) \geq 1 - \xi_i, \xi_i \geq 0; f(x) = \text{sign}(w \cdot x + b)$ where w is the weight vector defining the separating hyperplane, b is the bias, x_i and y_i are the gene expression profile and class label of sample i, ξ_i are slack variables allowing some points to fall within or across the margin, and C (cost) controls the trade-off between margin width and training error.		
- A fundamentally different mechanism, adding algorithmic diversity to the convergence analysis. - a linear rather than non-linear kernel was chosen because a linear decision boundary has lower complexity, reducing the risk of overfitting given this study's relatively small sample size.	Effective on GSE54456 (He et al., 2024) and a separate cohort (Liu et al., 2026; AUC 0.971).	- Requires $\log_2$ transformation and standardisation. The $\log_2$ transformation estimates nothing from the data and cannot leak; standardisation is data-estimated, so it was fit within each cross-validation fold on training-fold statistics only, preventing leakage (Section 3.5.4.2). - Cannot capture non-linear gene interactions. An accepted trade-off; low complexity reduces overfitting and adds algorithmic diversity to the analysis.

Table 3.6: Models considered but not selected, with the reason each was not used.

Model considered	Reason not used
Penalised logistic regression	Already represented through the elastic net used for feature selection (Section 3.5.4.2).
Naïve Bayes	Assumes conditional independence between features, strongly violated by the many correlated genes in this dataset (Li et al., 2014).
Neural networks	On tabular data, gradient-boosted tree ensembles outperform neural networks while being less computationally expensive (Yildiz & Kalayci, 2025).

3.5.4 Nested Cross-Validation

This study had to separate two jobs that are easy to merge together: building each model (choosing which genes to put into the model and tuning its hyperparameters) and estimating how well the finished model generalises to unseen samples. These two jobs must be carried out on different samples, for two related reasons. First, whenever a model's hyperparameters are chosen because they minimise error on a particular set of samples, part of that success reflects the model fitting the noise in those samples rather than a reproducible classification signal, so performance measured on those same samples is inflated (Varma & Simon, 2006). Second,

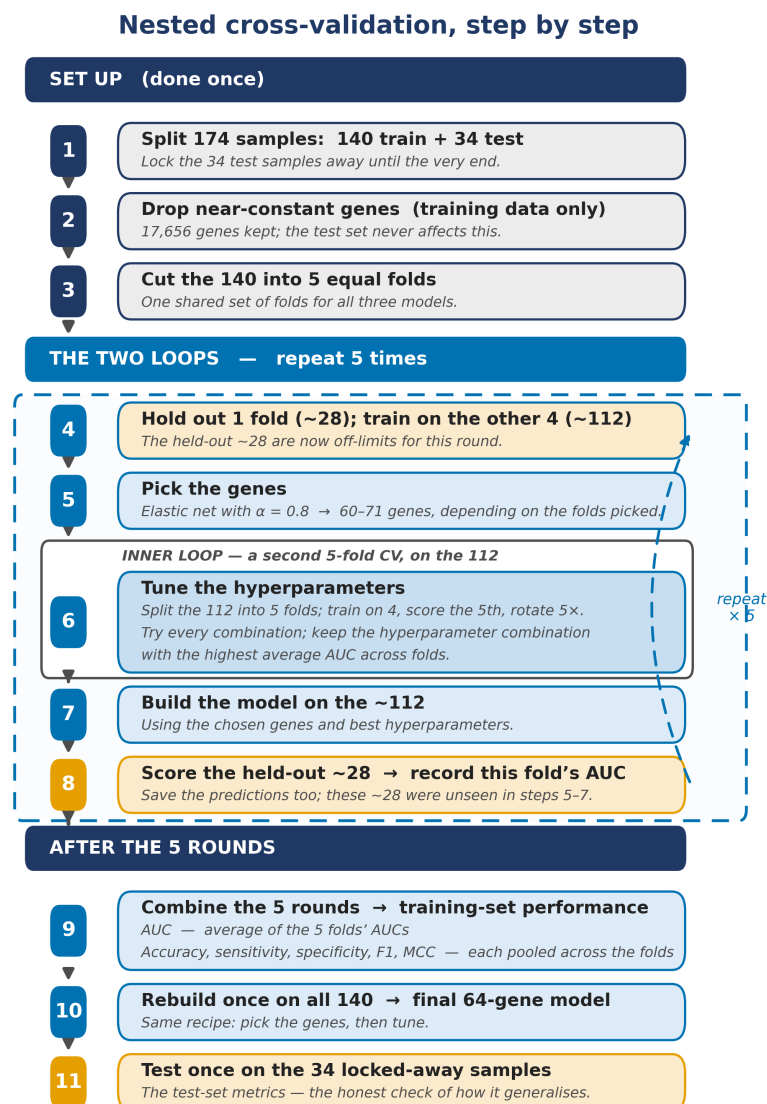
feature selection is itself a modelling decision. With genes vastly outnumbering samples, choosing the gene panel using the same samples that are later used to judge performance, rather than holding those samples back during feature selection, also overestimates performance (Ambroise & McLachlan, 2002). Both are forms of data leakage.

Two simpler ways of estimating performance within the training data were considered and rejected. Comparing them is what motivated the selection of the nested cross-validation approach for this study. A single train/test split is weakened by variance. It yields performance metrics from one arbitrary division, so the metrics depend heavily on which split is drawn, and this instability is pronounced at small sample sizes (Vabalas et al., 2019). Ordinary, non-nested cross-validation removes that instability by repeatedly partitioning the training set into folds and rotating which fold is held out but using those same held-out folds both to tune models and to report their performance makes performance estimates optimistic (Varma & Simon, 2006). Nested cross-validation keeps the stability while removing the bias. An outer loop repeatedly holds out one fold of the training set for performance estimation, and on the remaining outer training samples the model is selected (feature selection is performed on that iteration's outer-training samples, and an inner five-fold cross-validation then tunes the hyperparameters), so the held-out fold plays no part in either selection step. This is carried out entirely within the 140 training samples and provides reliable internal estimates of performance (Vabalas et al., 2019). The 34 held-out samples therefore remain completely unseen during model building and are used only as a final, independent check of how the model performs on samples withheld from the entire process.

The nested cross-validation design implemented in this study is shown in Figure 3.6. The 140 training samples were partitioned into five stratified outer folds once, using a fixed seed, so the same folds were reused across all three classification models (SVM, XGBoost, and random forest). In any given fold, each classifier saw exactly the same samples. Each classifier was then run through its own nested cross-validation, with its own inner tuning loop and its own final model. On each iteration of the outer loop, four of the five folds (approximately 112 samples) formed the outer-training set. Elastic net feature selection was applied once to this outer-training set (Section 3.5.4.1), producing a gene panel specific to that iteration. Because all three models shared the same folds, and elastic net used a fixed seed, each iteration therefore produced an identical gene panel across the three models. The models therefore differed only in the classification algorithm, so any performance differences reflected the algorithm rather than the samples or genes used. A five-fold cross-validated grid search was then run on the selected genes to tune each model's hyperparameters, retaining the combination with the

highest average area under the curve (AUC) across the inner folds (Section 3.5.4.2). The tuned model was then refit on the full outer-training set using the selected genes and was used to predict the held-out outer fold. This process was repeated until every fold had served once as the held-out fold. The AUC was then averaged across the five outer folds, while accuracy, sensitivity, specificity, F1, and the Matthews correlation coefficient were computed once on the pooled held-out predictions. Together, these formed the nested cross-validated (training-set) performance estimates. Finally, the complete procedure was refit on all 140 training samples to yield a single final model per algorithm, evaluated once on the 34-sample test set.

Figure 3.6: The nested cross-validation procedure. The setup steps were performed once. Steps 4–8 form the outer loop, repeated five times with a different fold held out in each outer iteration. Genes were selected once per outer iteration, and the inner loop (Step 6) tuned hyperparameters by a second five-fold cross-validation. The final model was refit on all 140 training samples and evaluated once on the held-out test set.



3.5.4.1 Elastic Net Feature Selection

After variance filtering, 17,656 genes remained against 140 training samples. Gene expression datasets exemplify the ‘large p, small n’ problem, in which the number of genes far exceeds the number of samples, a setting prone to overfitting (Khan et al., 2025). Therefore, elastic net regularisation was used for feature selection. It is a penalised regression that mixes the L1 (lasso) and L2 (ridge) penalties via a parameter α ($\alpha = 1$ gives a pure lasso, $\alpha = 0$ a pure ridge) (Zou & Hastie, 2005). This was preferred over a pure lasso, which cannot select more variables than there are samples and chooses one gene arbitrarily from a set of correlated genes while discarding the rest, and over a pure ridge, which performs no feature selection. The elastic net retains sparsity but, through its grouping effect, retains correlated informative genes together rather than dropping them arbitrarily (Zou & Hastie, 2005), which suits a compact yet biologically informative biomarker panel.

α was fixed at 0.8 following a diagnostic run on the full training set (140 samples), which varied α from 0.1 to 1.0 and recorded the panel size and cross-validated error at each value (Table 3.7). Error was minimised at $\alpha = 0.2$ (384 genes) but changed little across a broad range of α values. Raising α to 0.8 reduced the panel to 64 genes for only an approximately 0.003 increase in error. $\alpha = 0.8$ was therefore chosen to prioritise a smaller, more interpretable panel at a small cost to performance. A pure lasso was not chosen because setting $\alpha = 1.0$ (reducing to 19 genes) gave the highest cross-validation error of any setting tested (Table 3.7). Within nested cross-validation, the filter was refit at this fixed $\alpha = 0.8$ on each outer-training set, with λ chosen by the one-standard-error rule (Friedman et al., 2010). Because selection was repeated each iteration independently, the panel size varied from one iteration to the next (60–71 genes). The genes selected in each iteration are listed in Appendix A, the genes selected in all five outer iterations (the stable core) in Appendix B, and the final 64-gene panel, selected on the full 140-sample training set, in Appendix C. Because the elastic net used a fixed seed and shared folds, these selections were identical across the three classification models.

Table 3.7: Elastic net alpha diagnostic on the full training set. For each hyperparameter α , the table reports the number of genes selected at the one-standard-error λ and the corresponding cross-validated error. $\alpha = 0.2$ minimised the error, but $\alpha = 0.8$ (used in this study) reduced the panel to 64 genes at negligible additional cost; $\alpha = 1.0$ is a pure lasso.

Hyperparameter (α)	Number of Genes Selected	Cross-validation Error
0.1	675	0.03425
0.2 (CV-optimal)	384	0.02571
0.3	265	0.03327
0.4	197	0.02762
0.5	154	0.02744
0.6	121	0.02886
0.7	93	0.02857
0.8 (used)	64	0.02856
0.9	45	0.04123
1.0	19	0.05694

3.5.4.2 Feature Standardisation and Hyperparameter Tuning

Standardisation was applied selectively rather than uniformly, depending on the algorithm used. The elastic net feature selection filter operates on standardised predictors (Zou & Hastie, 2005), so standardisation was built into its fitting. The support vector machine was also standardised. Because the margin depends on distances in feature space, it is sensitive to feature scale and requires predictors on comparable ranges (Hsu et al., 2003). The tree-based models were left unstandardised, being scale-invariant (Table 3.5). Where standardisation was used, z-score standardisation was chosen over min–max scaling because gene expression data contain extreme but legitimate values; min–max scaling is sensitive to such outliers, as it compresses values into a fixed range, whereas z-score standardisation uses the mean and standard deviation and so is less distorted by extreme values (Laska & Yolanda, 2024). In every case, the scaling parameters were estimated from the training data only, so samples in the held-out cross-validation folds and in the test set were scaled using estimates they had no part in defining, preventing data leakage (Pinheiro et al., 2025).

Each model's hyperparameters were tuned by a grid search within the inner cross-validation loop, selecting the combination with the highest mean AUC across the inner folds. Grids were deliberately restricted to the hyperparameters known to most influence performance. The values searched and rationale are given in Table 3.8, and the final values selected for each model (refit on the full 140-sample training set) in Table 3.9. The number of trees for the random forest model was fixed at 1,000 rather than tuned. Because each tree is grown independently, adding trees does not cause overfitting in the way boosting rounds do,

and the number of trees is better set to a large, computationally manageable value than treated as a tuning parameter (Probst & Boulesteix, 2017). Across the three models, tuning generally favoured the simplest settings. This is consistent with the highly separable classification task in this study, where the main risk is overfitting a relatively small sample, rather than underfitting. The exception was the random forest's minimum node size, where the smallest value was selected. Because the random forest averages over many decorrelated trees (Breiman, 2001), deeper individual trees do not carry the overfitting risk that they would in a single decision tree.

Table 3.8: *Hyperparameters tuned by grid search within the inner cross-validation loop, the values searched, and the rationale for each. Hyperparameters were selected by the highest mean AUC across the inner cross-validation folds. Fixed (untuned) settings are noted beneath the table.*

Model	Hyperparameter	Values searched	Rationale
XGBoost	Maximum tree depth	2, 3	Shallow depths (2–3) were searched to keep individual trees simple, since deeper trees can fit noise in the training data and overfit (Maniwa et al., 2024).
XGBoost	Number of Boosting Rounds	50, 100, 150	The search spanned small to moderate ensembles (50–150 boosting rounds). Too few rounds can underfit, while too many can overfit (Natekin & Knoll, 2013), so the range was kept deliberately small.
XGBoost	Learning rate (eta)	0.1, 0.3	Lower values apply shrinkage, reducing each tree's contribution and thereby the risk of overfitting (Chen & Guestrin, 2016). A conservative rate (0.1) and a higher one (0.3) were therefore compared.
XGBoost	Column subsampling	0.6, 0.8	Every tree was built on a random subsample of the features, 60% or 80% of the genes, rather than the full set, so that the trees differed from one another. Chen and Guestrin (2016) include column subsampling among the techniques that reduce overfitting. The two values bracket a useful range: 0.6 produces more varied trees, while 0.8 gives each tree access to more of the genes.
Random forest	Variables per split (mtry)	5, 10, 20, 30	Among the random forest hyperparameters, mtry has the greatest influence on predictive performance (Probst et al., 2019). Small values (5) give diverse but individually weaker trees, while large values (30) give stronger but more similar trees, so the range was chosen to balance diversity against individual tree strength.

Model	Hyperparameter	Values searched	Rationale
Random forest	Minimum node size	5, 10, 15	Node size sets how many observations a terminal node must retain, which controls how far each tree is grown: lower values let the trees keep splitting into deep, fine-grained structure, while higher values halt splitting sooner and give shallower, simpler trees (Probst et al., 2019). Both ends of the range were searched so the model could settle wherever the balance between tree expressiveness and simplicity fell for this dataset.
SVM (linear)	Cost (C)	0.25, 0.5, 1, 2, 4	C controls the trade-off between margin width and tolerance for training errors (Hsu et al., 2003). Lower values impose stronger regularisation and may underfit, whereas higher values place greater emphasis on minimising training errors and may increase the risk of overfitting (Pahikkala et al., 2005). The range 0.25–4 was therefore selected to span comparatively stronger and weaker regularisation.

Fixed settings: *XGBoost*: minimum split loss (γ) = 0, minimum child weight = 1, row subsampling = 0.8; *random forest*: number of trees = 1,000 and splitting rule = Gini. all unlisted hyperparameters were left at their *RStudio* package defaults.

Table 3.9: Final hyperparameter values selected by nested cross-validation and used to refit each model on all 140 training samples.

Model	Final hyperparameters
XGBoost	nrounds = 50, maximum depth = 2, learning rate = 0.1, column subsampling = 0.6, row subsampling = 0.8, minimum split loss = 0, minimum child weight = 1
Random forest	mtry = 5, minimum node size = 5, splitting rule = Gini, number of trees = 1,000
SVM (linear)	C = 0.25

3.5.5 Performance Metrics

Model performance was evaluated using six performance metrics. No single metric captures every aspect of classification, so a combination was used to give a more complete assessment of performance. The metrics, their formulas, and the justification for the use of each of them is discussed in Table 3.10 below.

Table 3.10: Classification performance metrics used to evaluate the models, their formulas, and the justification for each. TP, FP, TN and FN denote true and false positives and negatives.

Metric	Formula	Justification
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall proportion of correct predictions, but it can be misleading under class imbalance (Tharwat, 2021); as this dataset is mildly imbalanced (92 psoriasis vs 82 normal), accuracy is reported alongside the other metrics rather than relied on alone.
Sensitivity (Recall)	$TP / (TP + FN)$	Proportion of psoriasis cases correctly identified. It is clinically important for minimising missed cases (Sokolova & Lapalme, 2009).
Specificity	$TN / (TN + FP)$	Proportion of normal samples correctly identified. It measures the control of false positives (Sokolova & Lapalme, 2009).
F1 score	$2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$	Harmonic mean of precision ($TP/(TP+FP)$) and recall; a high F1 requires the model to control both false positives and false negatives (Powers, 2020).
AUC (ROC) Area Under the Curve (Receiver Operating Characteristic)	True positive rate against false positive rate across all thresholds.	Ranges from 0 to 1, where 0.5 indicates performance no better than chance and 1.0 indicates perfect discrimination between the tissue types. Considers all possible decision thresholds and is robust to class imbalance (Li, 2024). Used as the primary model selection metric during hyperparameter tuning.
MCC (Matthews Correlation Coefficient)	$(TP \cdot TN - FP \cdot FN) / \sqrt{[(TP+FP)(TP+FN)(TN+FP)(TN+FN)]}$	Single score from -1 to +1, high only when both classes are classified well with few errors of either type, so it cannot be inflated by the majority class alone (Chicco & Jurman, 2020). Evaluates the final model at the 0.5 decision threshold, complementing the threshold-independent AUC.

For each model, these metrics, along with confusion matrices, and ROC curves were produced on both the nested cross-validated (training-set) predictions and the held-out test set. Bootstrap confidence intervals were computed around each AUC to indicate the precision of the estimate.

3.5.6 Model Feature Importance with SHAP

This interpretation step directly addresses the research gap identified in Section 2.3, where prior psoriasis studies applied post-hoc explanation to a single model, if at all. Here, SHAP (Shapley Additive explanations) was instead applied to all three classifiers on the 34-sample test set, identifying the genes driving each model's predictions and comparing these rankings across models to find the genes on which the algorithms agreed. Because correlation between genes distorts the importance a single model assigns (Tološi & Lengauer, 2011), and different algorithms are known to select different features as important (Saarela & Jauhiainen, 2021), a gene ranked highly by several model types is less likely to reflect any single algorithm's bias, making agreement across models more reliable than one model's explanation alone. For each prediction, SHAP assigns every gene a value and a direction indicating whether its contribution shifts the prediction towards psoriasis or normal (Lundberg & Lee, 2017).

SHAP was chosen over other post-hoc explanation methods. Permutation importance yields only a single global importance value per gene, with no indication of the direction of a gene's effect (whether it pushed predictions towards psoriasis or normal) or how that effect varies across samples. SHAP instead assigns every gene a signed value (positive or negative) for every individual prediction, so each gene's global importance, its direction of effect, and its per-sample contribution can all be recovered. Direction is informative because it allows each gene's predictive effect to be checked against its differential expression. For example, a gene upregulated in psoriatic skin should, if it carries a legitimate predictive signal, drive predictions towards psoriasis. This concordance is particularly informative for candidate genes with no recorded disease association with psoriasis, where such internal consistency is the main available evidence that the gene reflects a real predictive signal rather than noise. The per-sample detail supports this further, showing whether a gene's effect is consistent across the cohort or driven by a few samples, a distinction that global importance alone would not show. Local Interpretable Model-agnostic Explanations (LIME) was also considered but rejected because it fits a separate model for each prediction (Ribeiro et al., 2016). SHAP was therefore used to allow consistent comparison of gene contributions across predictions.

SHAP was computed using the R package providing the estimator appropriate to each model type. Exact Tree SHAP was used for the XGBoost and random forest models (Lundberg et al., 2020). The SVM is not tree-based, so Tree SHAP does not apply; the model-agnostic Kernel SHAP, which can be applied to any algorithm (Lundberg & Lee, 2017), was used instead. Because Kernel SHAP approximates Shapley values by sampling from the many

possible subsets of the 64 genes, and in this dissertation, it did not reach convergence for any of the 34 test samples, so the SVM's SHAP values were treated as approximate rather than exact. The direction of each gene's effect nonetheless remained stable. For each top-ten gene, expression level and SHAP value were strongly correlated ($|r| \geq 0.97$), so directions were interpreted with confidence while exact ranks and magnitudes were treated with caution.

For each model, genes were ranked by mean absolute SHAP value, and that model's ten highest-ranked genes were shown in two complementary graphs. A bar chart displayed each gene's global importance (its mean absolute SHAP value across all test samples) showing how much the gene influenced predictions overall. A beeswarm plot then showed the SHAP value for every individual sample, with horizontal position giving the magnitude and direction of each gene's effect and colour showing its expression level. Together, these convey a gene's overall importance, the direction of its effect, and how that effect relates to its expression, and were used to interpret each model individually. Across models, each gene was then recorded by how many models placed it among their ten most influential, indicating the breadth of support for it. This membership-level comparison is more robust than comparing magnitudes or exact ranks, which are distorted by the differing SHAP scales (log-odds for XGBoost; probability for random forest and SVM) and by the SVM's approximation. Because membership depends only on whether a gene reached the top ten, not on its precise position, it tolerates more estimation error than exact ranking. The remaining uncertainty is greatest for the SVM, where genes ranked near the top-ten boundary are the least reliably placed.

3.6 Candidate Biomarkers and Their Associated Diseases

Utilising DISGENET (RQ3)

A gene supported by only one line of evidence may reflect the bias of that particular method, so RQ3 combined evidence from the earlier research questions into a single set of criteria to identify candidate biomarkers supported by multiple different lines of evidence. Each criterion tests a distinct property: differential expression establishes that a gene's expression differs between psoriatic and normal skin samples, cross-validation selection stability establishes that a gene is chosen robustly rather than the result of one data split, and SHAP importance establishes that at least one trained model relied on it. For each gene, four quantities were recorded: its cross-validation iteration stability (0–5), its differential expression and direction, its mean absolute SHAP value and rank in each model, and whether it fell in each

model's top ten. A gene was nominated as a candidate biomarker only if it met all three criteria listed in Table 3.11.

Table 3.11: *Pre-specified criteria for nominating a candidate biomarker. A gene had to satisfy all three.*

	Criterion	Threshold
(a)	Differentially expressed	The gene meets FDR-adjusted $p < 0.05$ and $ \log_2FC > 1$
(b)	Cross-Validation Selection stability	The gene is retained in all five cross-validation outer iterations (See Appendix B)
(c)	Model importance (SHAP)	The gene ranks within the top ten by mean absolute SHAP value in at least one classification model.

DISGENET, a large public database aggregating gene-disease associations (MedBioinformatics Solutions, 2026a), was then queried for each nominated gene as external annotation rather than as a selection criterion. For each gene, the association score for psoriasis and its three most strongly associated diseases were recorded, distinguishing candidates already linked to psoriasis from potentially novel ones. The fields reported and defined in Table 3.12 follow the DISGENET metrics documentation (MedBioinformatics Solutions, 2026b).

Table 3.12: *DISGENET fields reported for each candidate gene.*

Metrics	Definition
GDA Score (Gene-disease Association Score)	DISGENET gene–disease association score (0–1). It reflects how well-supported an association is, weighted by the type and reliability of the evidence (curated and expert-reviewed sources contribute more than text-mined literature) rather than by the number of publications alone. Higher values indicate a stronger supported association.
DSI (Disease Specificity Index)	Disease Specificity Index (0.23–1). A gene-level measure of how specific a gene is to diseases. A higher DSI indicates the gene is associated with fewer diseases (more specific). A lower DSI indicates association with many diseases.
Number of PMIDs (Number of PubMed Papers)	The number of supporting publications (PubMed IDs) for the gene–disease association. More publications indicate stronger evidential support.
Top Three Associated Diseases	The three diseases with the highest GDA score for each gene, indicating the other conditions it has been associated with.

3.7 Software and Implementation

All analyses were implemented in RStudio (version 2025.09.0+387). The packages used are listed in Table 3.13. For reproducibility, a random seed (42) was set for all data partitioning, fold generation, and model fitting, and the elastic net filter used a fixed internal seed (1000) so that gene selection was identical across the three model branches within a given outer iteration. The RStudio package xgboost was deliberately set at version 1.7.8.1, as later releases are incompatible with the caret training interface.

Table 3.13: *RStudio software packages used in the analysis, with the version listed and purpose.*

Package	Version	Purpose
<i>limma</i>	—	Differential expression
<i>glmnet</i>	—	Elastic-net feature selection
<i>nestedcv</i>	—	Nested cross-validation
<i>caret</i>	—	Model training, tuning, and confusion-matrix metrics
<i>xgboost</i>	1.7.8.1	Gradient-boosted classifier
<i>ranger</i>	—	Random forest classifier
<i>kernlab</i>	—	SVM classifier (linear, via caret)
<i>shapviz, treeshap</i>	—	Tree SHAP explanations (tree models)
<i>kernelshap</i>	—	Kernel SHAP explanations (SVM)
<i>pROC</i>	—	ROC / AUC with bootstrap CIs
<i>mltools</i>	—	Matthews correlation coefficient

3.8 Ethical Considerations

This study is a secondary analysis of publicly available and fully anonymised data. The GSE54456 samples carry only anonymous alphanumeric identifiers, with no personal, demographic, or clinical information, so re-identification is not a risk. Informed consent and ethical approval for the original biopsies were obtained by the original investigators (Li et al., 2014). No new tissue was collected and no participants were contacted. In accordance with University of Sheffield policy, a University Research Ethics Committee-approved self-declaration was completed, confirming that the research involved only robustly anonymised existing data and the project was approved (reference 074624; see Appendix D).

A brief reflection on the scope of these findings is warranted. The results presented in the next chapter are specific to this cohort. Therefore, the nominated biomarkers are candidates requiring validation in independent cohorts before any clinical claim can be made.

The following chapter presents and discusses the results of this study in relation to each research question.

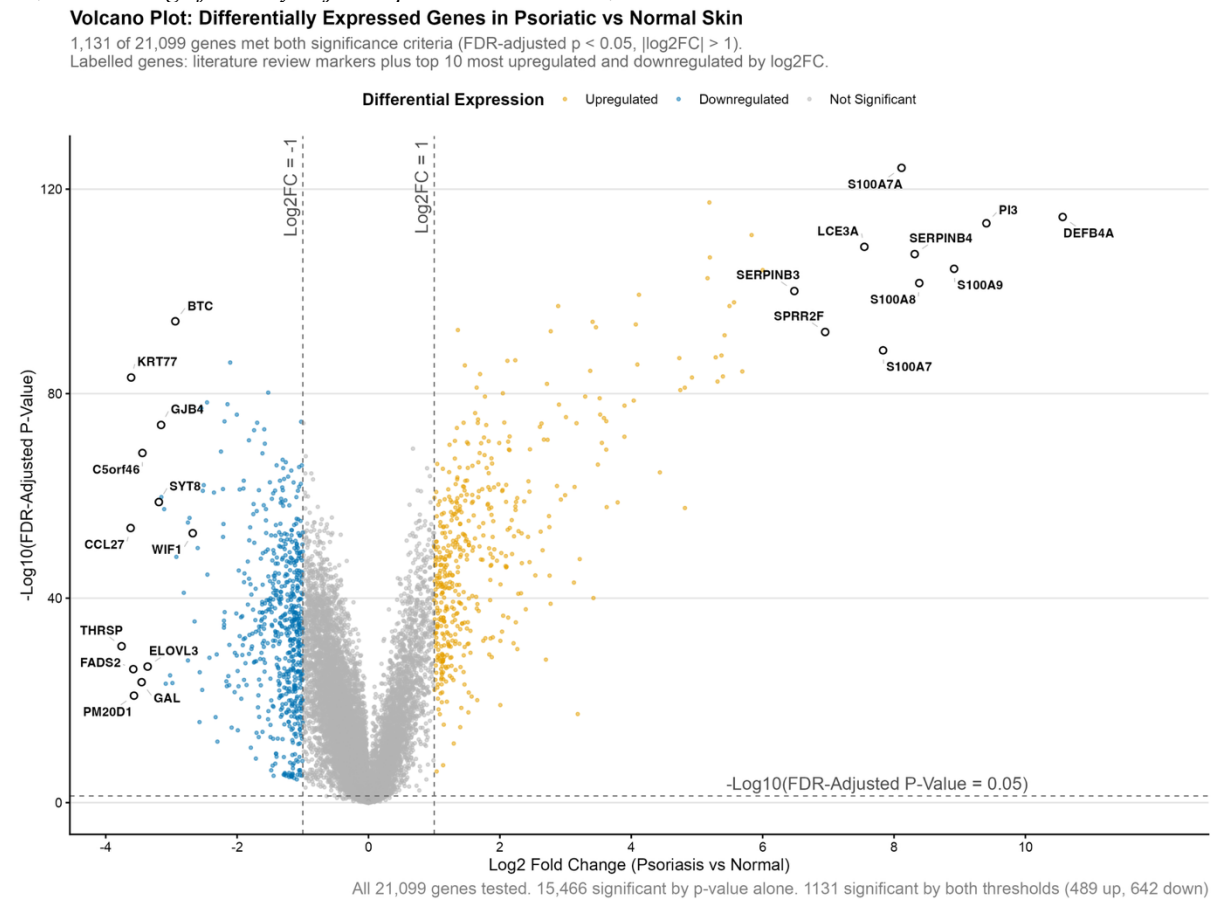
Chapter 4: Results

This chapter presents the results produced by the methods set out in Chapter 3, addressing each research question in turn. It first reports the differentially expressed genes between psoriatic and non-psoriatic skin (Section 4.1, RQ1), then the classification performance of the three models and their agreement on the most important genes (Section 4.2, RQ2), and finally the candidate biomarkers and their disease associations (Section 4.3, RQ3).

4.1 Differential Expression (RQ1)

Differential expression was assessed with limma across all 21,099 retained genes (Section 3.4). At a false discovery rate of 5%, 15,466 genes were significant on the FDR-adjusted p -value alone. Applying the additional biological-effect threshold of $|\log_2\text{FC}| > 1$ reduced this to 1,131 genes meeting both criteria. This was comprised of 489 upregulated and 642 downregulated in psoriatic relative to normal skin. The volcano plot in Figure 4.1 shows this separation. The most extreme upregulated gene reached a $\log_2$ fold change above 10, whereas the downregulated set was more modest in magnitude, extending to roughly -3.8 .

Figure 4.1: Volcano plot of differential expression between psoriatic and normal skin. Each point is one of the 21,099 tested genes, plotted by $\log_2$ fold change (x-axis; positive = upregulated in psoriasis) against $-\log_{10}$ FDR-adjusted p -value (y-axis; the $-\log_{10}$ transformation is applied for visualisation only, spreading out the smallest p -values, and does not affect the significance calls). Dashed lines mark the significance thresholds ($|\log_2FC| > 1$ and FDR-adjusted $p < 0.05$). Orange points are the 489 significantly upregulated genes and blue the 642 downregulated. Grey points are not significant. Labelled genes are the genes mentioned in Section 2.2.1 of the literature review and the top 10 upregulated and downregulated genes by fold change. Of the 21,099 genes, 15,466 were significant by adjusted p -value alone and 1,131 met both thresholds.



The ten genes with the largest positive and negative fold changes are reported in Tables 4.1 and 4.2. The most upregulated genes were led by *DEFB4A* ($\log_2FC = 10.57$), followed by *PI3* and several *S100* family members, all with extremely small adjusted p -values ($< 10^{-89}$). The downregulated set was led by *THRSP*, *CCL27* and *KRT77*. The complete differential expression results for all 21,099 genes, and the full lists of significantly upregulated and downregulated genes ($|\log_2FC| > 1$ and FDR-adjusted $p < 0.05$), are provided as supplementary CSV files on the Google Drive.

Table 4.1: The ten most upregulated genes in psoriatic relative to normal skin, ranked by $\log_2$ fold change. All met both significance criteria (FDR-adjusted $p < 0.05$ and $|\log_2FC| > 1$).

Gene	Log ₂ FC	Adjusted p-value
DEFB4A	10.57	2.81×10^{-115}
PI3	9.41	4.81×10^{-114}
S100A9	8.91	3.82×10^{-105}
S100A8	8.38	2.37×10^{-102}
SERPINB4	8.31	5.05×10^{-108}
S100A7A	8.11	6.88×10^{-125}
S100A7	7.83	3.44×10^{-89}
LCE3A	7.55	1.94×10^{-109}
SPRR2F	6.95	8.78×10^{-93}
SERPINB3	6.48	8.52×10^{-101}

Table 4.2: The ten most downregulated genes in psoriatic relative to normal skin, ranked by $\log_2$ fold change (most negative first). All met both significance criteria.

Gene	Log ₂ FC	Adjusted p-value
THRSP	-3.76	2.59×10^{-31}
CCL27	-3.62	1.97×10^{-54}
KRT77	-3.61	7.00×10^{-84}
FADS2	-3.58	7.75×10^{-27}
PM20D1	-3.57	1.17×10^{-21}
GAL	-3.45	2.78×10^{-24}
C5orf46	-3.44	4.11×10^{-69}
ELOVL3	-3.36	2.44×10^{-27}
SYT8	-3.19	1.61×10^{-59}
GJB4	-3.16	1.34×10^{-74}

4.2 Classification Performance and Cross-Model Agreement (RQ2)

All three models classified the two tissue types without error, under both nested cross-validation and held-out testing. On the 140 training samples, nested cross-validation returned 1.00 on all six performance metrics for every model, with an AUC standard deviation of zero across the five outer folds (Table 4.3). The same perfect performance held on the independent 34-sample test set. The confusion matrices (Table 4.4) confirm this. No sample was misclassified by any model, in either the cross-validation or the test evaluation. This uniform performance is consistent with the clear separation already visible in the exploratory analysis, where PC1 alone separated psoriatic from normal samples (Section 3.3.5), and with the strong within-cohort separability reported for psoriasis in the literature (Section 2.2.2). Because feature selection and tuning were performed only on each iteration's training samples and never

on the held-out fold, these estimates were not inflated by information leakage (Section 3.5.4), and the matching test-set performance supports this. Performance is reported here to confirm that each model classified reliably, a prerequisite for the feature-importance analysis.

Table 4.3: Classification performance for each model, under nested cross-validation on the 140 training samples and on the held-out 34-sample test set. Cross-validated AUC is the mean across the five outer folds (standard deviation in parentheses); all other cross-validated metrics were computed on the pooled held-out predictions, and test-set metrics on the 34 held-out samples. All metrics reached 1.000 for every model under both sets.

Metric	XGBoost CV	XGBoost Test	Random Forest CV	Random Forest Test	SVM CV	SVM Test
AUC	1.000 (0.000)	1.000	1.000 (0.000)	1.000	1.000 (0.000)	1.000
Accuracy	1.000	1.000	1.000	1.000	1.000	1.000
Sensitivity	1.000	1.000	1.000	1.000	1.000	1.000
Specificity	1.000	1.000	1.000	1.000	1.000	1.000
F1	1.000	1.000	1.000	1.000	1.000	1.000
MCC	1.000	1.000	1.000	1.000	1.000	1.000

Table 4.4: Confusion matrices for each model under cross-validation and on the test set. Columns give the four cells of the two-class matrix, expressed as predicted → actual counts. There were no misclassified samples. Every sample was correctly assigned to its tissue type under both cross-validation and on the test set.

Model	Evaluation set	Normal → Normal	Psoriasis → Normal	Normal → Psoriasis	Psoriasis → Psoriasis
XGBoost	Cross-validation	66	0	0	74
XGBoost	Test set	16	0	0	18
Random Forest	Cross-validation	66	0	0	74
Random Forest	Test set	16	0	0	18
SVM	Cross-validation	66	0	0	74
SVM	Test set	16	0	0	18

The corresponding receiver operating characteristic (ROC) curves are shown for the cross-validation set in Figure 4.2 and for the test set in Figure 4.3. In both, all three curves track the top-left corner, giving areas under the curve of 1.000 (95% CI 1.000–1.000).

Figure 4.2: ROC curves for the three models under nested cross-validation on the 140 training samples. True positive rate (sensitivity) is plotted against false positive rate ($1 - \text{specificity}$). All three curves reach the top-left corner, corresponding to an AUC of 1.000 (95% CI 1.000–1.000) for each model. The diagonal marks chance performance (AUC = 0.5).

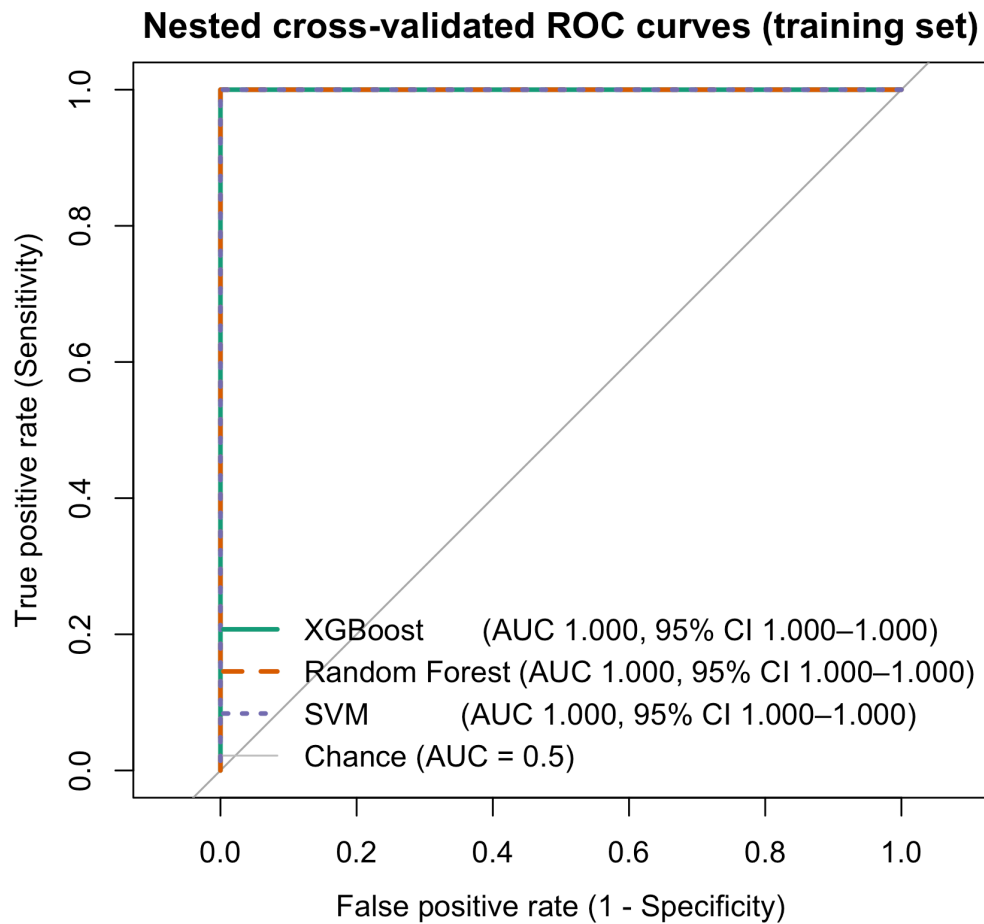
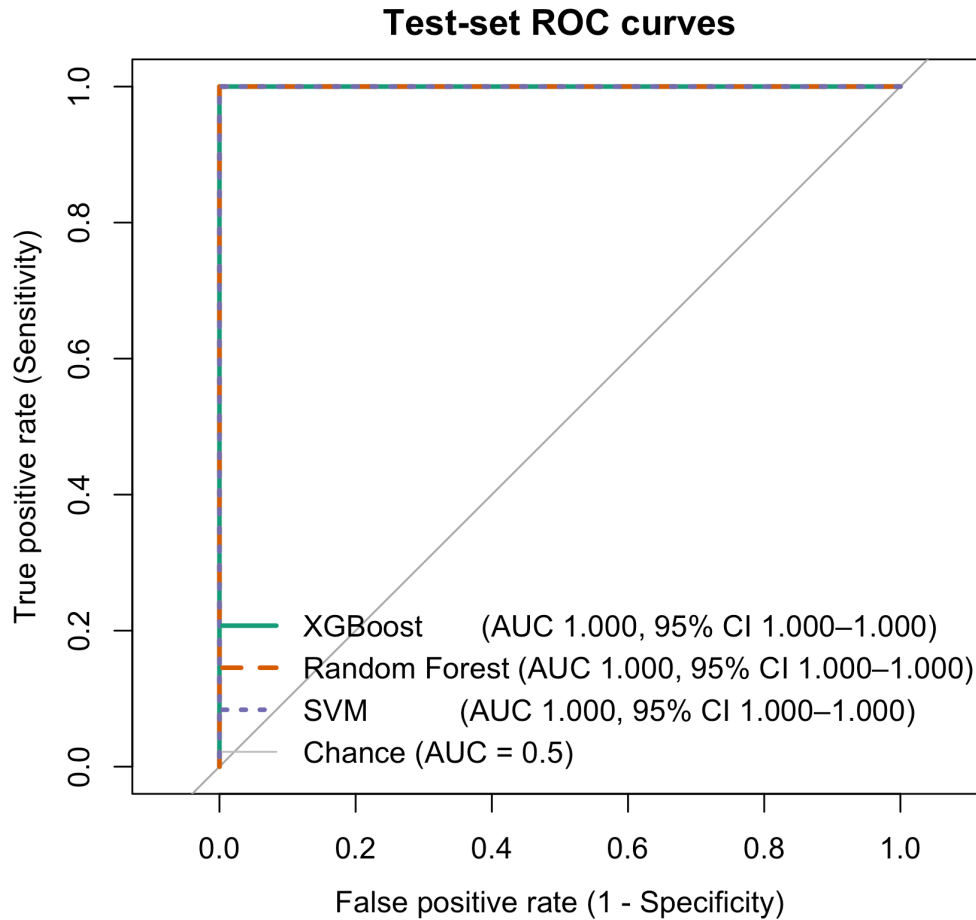


Figure 4.3: ROC curves for the three models on the 34-sample test set. Every model achieved an AUC of 1.000 (95% CI 1.000–1.000), confirming that the classifiers perfectly separated the two tissue types on unseen samples drawn from the same cohort.



SHAP was used to identify the genes each model relied upon the most, applied to all three classifiers on the 34-sample test set (Section 3.5.6). For each model, a bar chart shows global importance (mean absolute SHAP value) for the ten highest-ranked genes, and a beeswarm plot shows the corresponding per-sample SHAP values coloured by gene expression level. Two features of the SHAP output must be kept in mind when reading the figures. First, the models operate on different output scales. XGBoost SHAP values were on the log-odds scale, whereas the random forest and SVM values were on the probability scale, so magnitudes are not directly comparable across models. Cross-model comparison therefore relies on the direction of each gene's effect and on whether a gene appears among a model's ten most important, rather than on exact ranks. Second, the models differed markedly in how many genes they used. XGBoost assigned non-zero importance to only seven of the 64 panel genes, the random forest to 56, and the SVM to all 64. Because the SVM's Kernel SHAP estimates did not converge, its exact ranks and magnitudes are treated with caution, though its directions remained stable ($|r| \geq 0.97$ for all top ten genes).

The three models' top-ranked genes overlapped only on *ABCG4* (XGBoost 1st, random forest 3rd, and SVM 9th). Every other gene was specific to a single model. As expected under correlated features (Section 2.2.2), the algorithms largely favoured different genes, which makes *ABCG4*'s appearance in the top ten of all three models notable. Its directional effect was equally consistent: across all three models, higher *ABCG4* expression drove predictions towards psoriasis (Figures 4.5, 4.7, and 4.9), matching its upregulation in the differential expression analysis (Section 4.1).

Figure 4.4: XGBoost SHAP mean absolute importance for the top ten genes (test set, $n = 34$). Bars give each gene's mean absolute SHAP value across all test samples on the log-odds scale. Only 7 of the 64 panel genes received non-zero importance, with *ABCG4* (1.918) and *AKR1B10* (1.287) dominating.

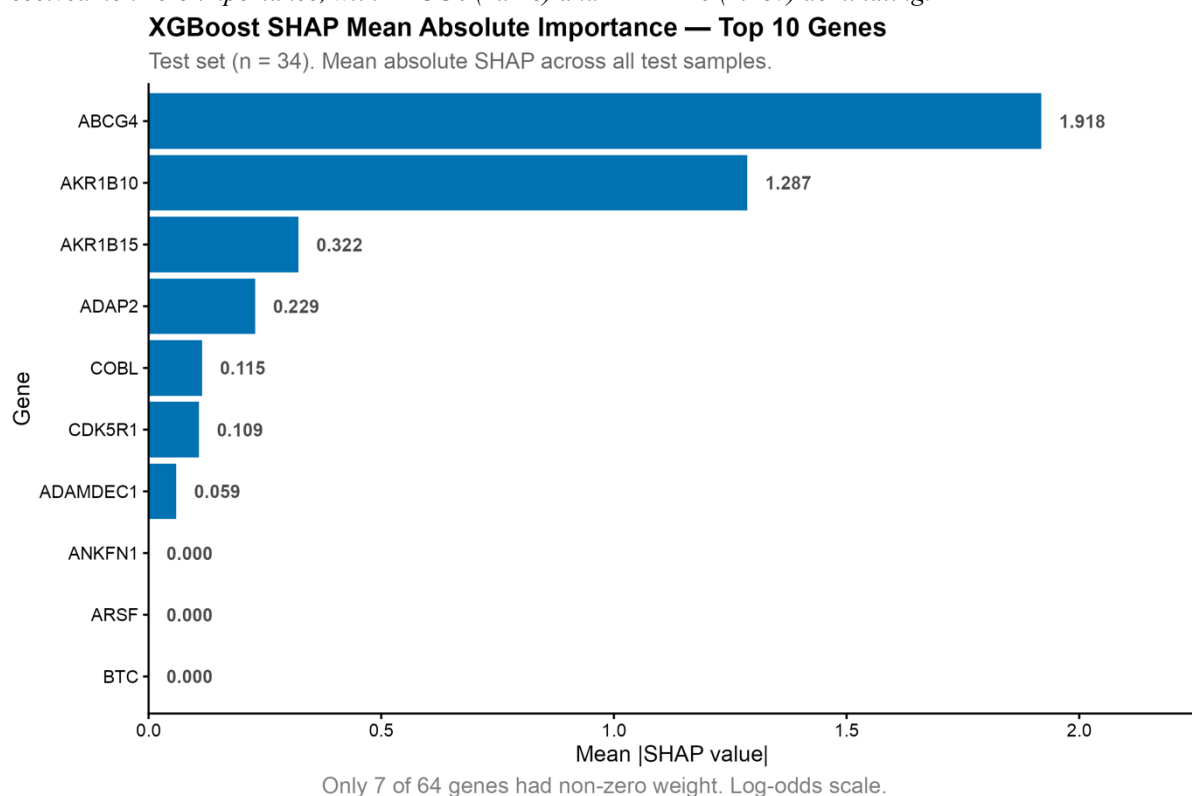


Figure 4.5: XGBoost SHAP beeswarm plot for the top ten genes (test set, $n = 34$). Each point is one sample; horizontal position gives the SHAP value (positive = towards Psoriasis, negative = towards Normal) and colour is the gene's expression level (orange = high, blue = low). For ABCG4 and AKR1B10, high expression pushed predictions towards psoriasis.

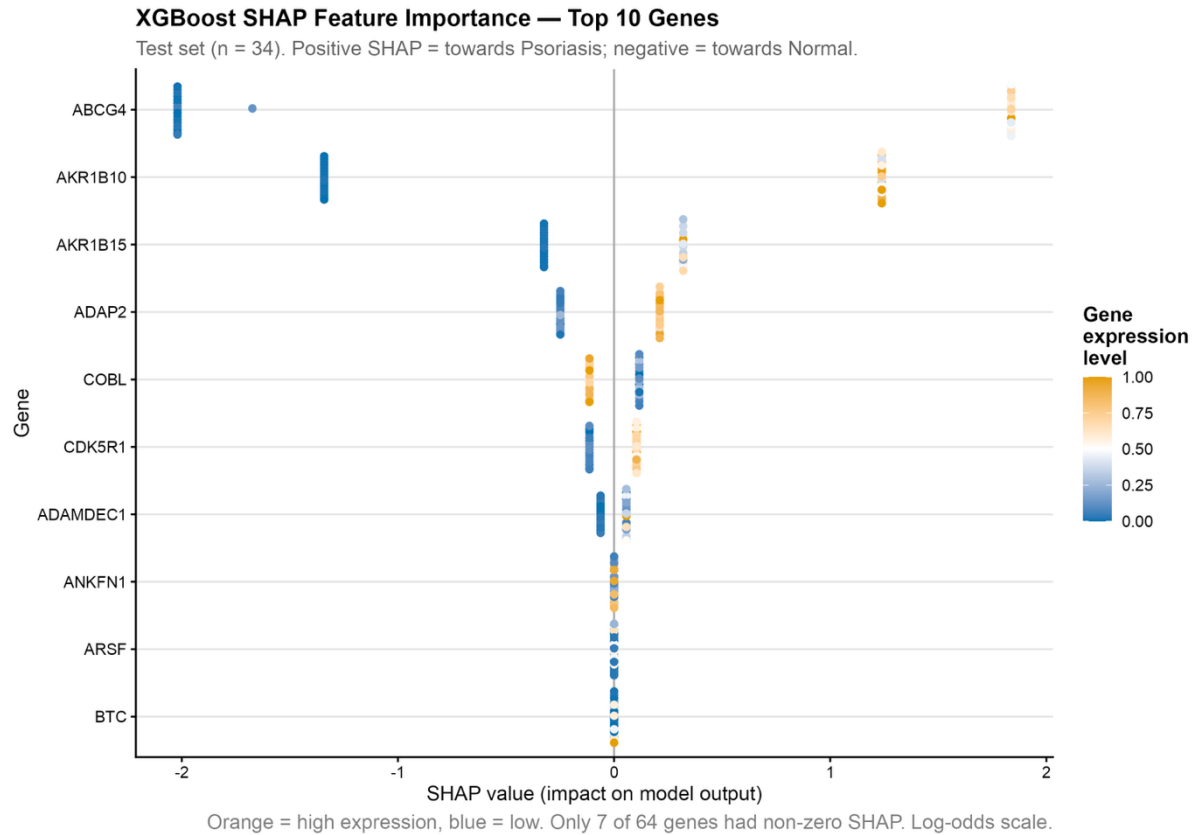


Figure 4.6: Random forest SHAP mean absolute importance for the top ten genes (test set, $n = 34$), on the probability scale. *VNN3*, *S100A8* and *ABCG4* occupied the top three positions, with importance values closely spaced across the ten genes. Of the 64 panel genes, 56 received non-zero importance.

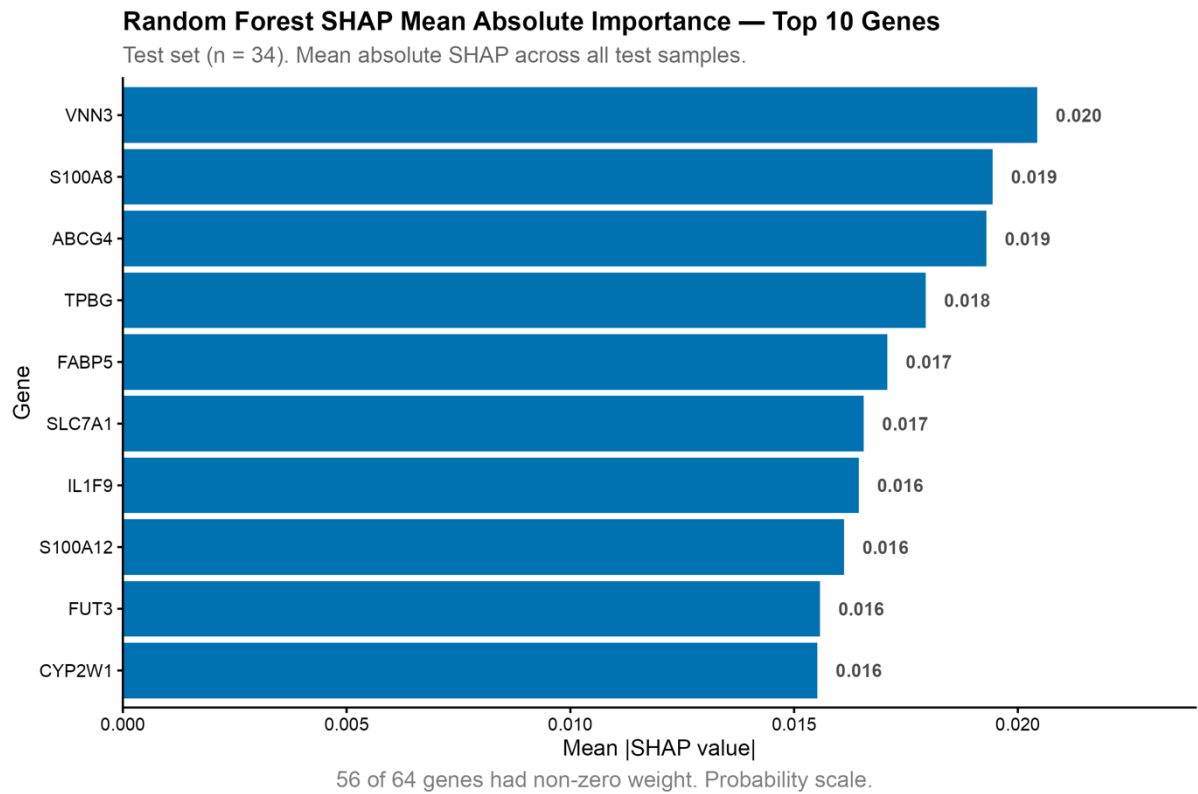


Figure 4.7: Random forest SHAP beeswarm plot for the top ten genes (test set, $n = 34$). Each point is one sample; horizontal position gives its SHAP value (positive = towards psoriasis, negative = towards normal) and colour is the gene's expression level (orange = high, blue = low). For most of the top 10 genes, higher expression tended to push predictions towards psoriasis and lower expression towards normal, though the strength and clarity of this relationship varied across genes.

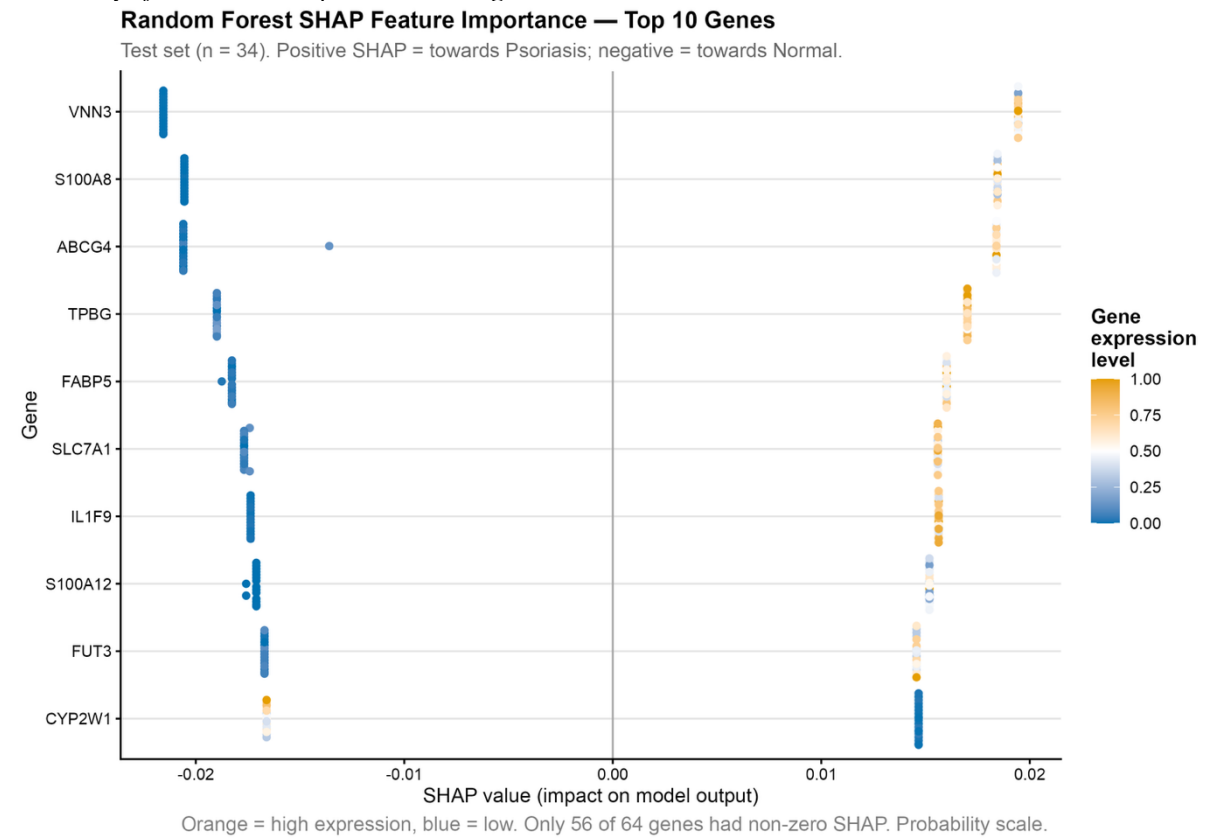


Figure 4.8: Support vector machine SHAP mean absolute importance for the top ten genes (test set, $n = 34$), on the probability scale. TAP1 was ranked highest, followed by a closely spaced set of genes. Because the Kernel SHAP estimates did not converge, the exact ranks and magnitudes are approximate. All 64 genes received non-zero attribution, an inherent property of Kernel SHAP.

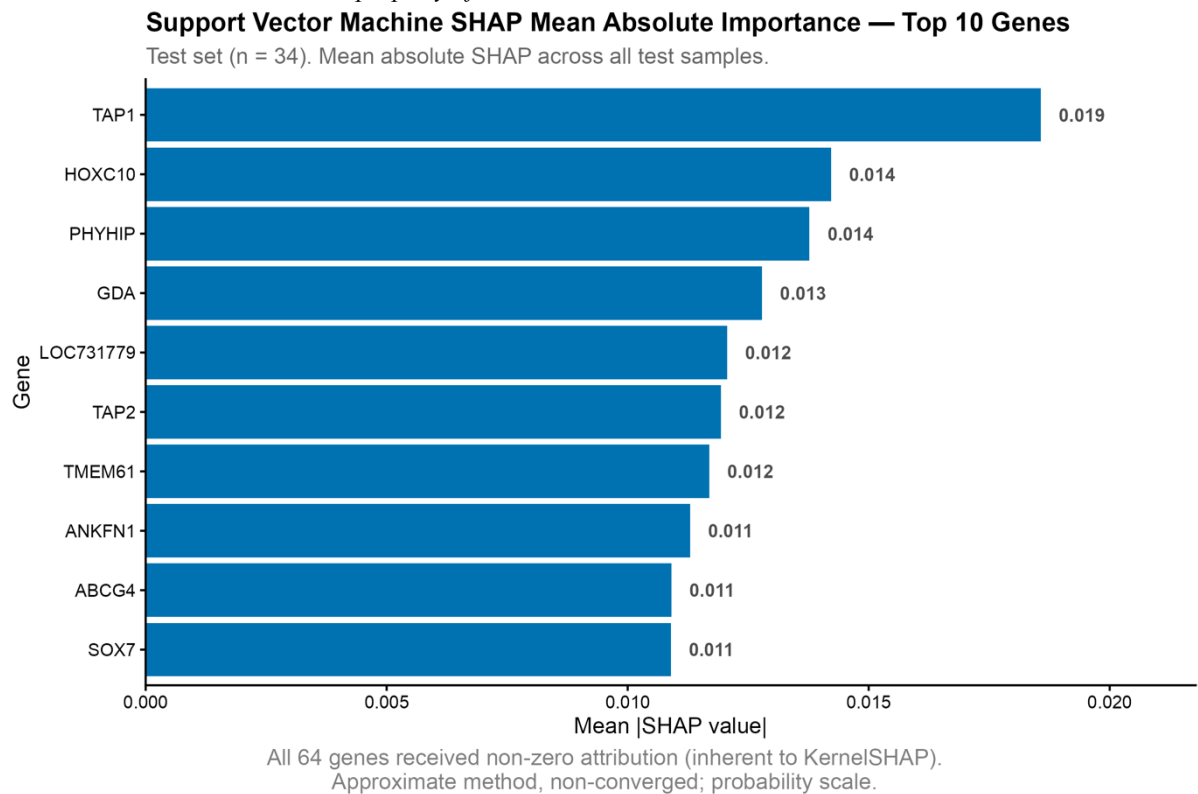
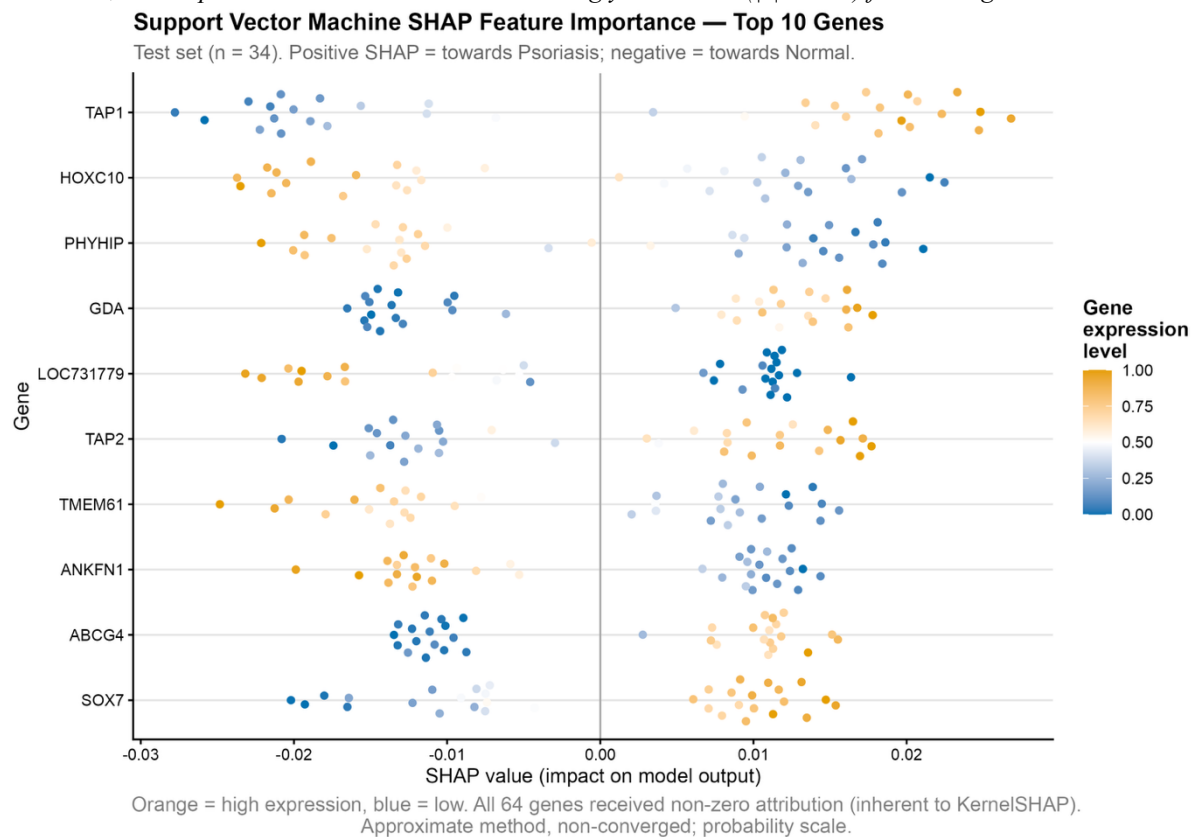


Figure 4.9: Support vector machine SHAP beeswarm plot for the top ten genes (test set, $n = 34$). Each point is one sample; horizontal position gives its SHAP value (positive = towards psoriasis, negative = towards normal) and colour is the gene's expression level (orange = high, blue = low). Because the Kernel SHAP estimates did not converge, magnitudes and exact ranks are approximate. However, the direction of each gene's effect was consistent, with expression level and SHAP value strongly correlated ($|r| \geq 0.97$) for all ten genes.



4.3 Candidate Biomarkers and Disease Associations

The three lines of evidence were combined using the pre-specified criteria in Table 3.11: differential expression meeting FDR-adjusted $p < 0.05$ and $|\log_2FC| > 1$, retention by the elastic-net filter in all five cross-validation outer iterations (Appendix B), and placement within the top ten mean absolute SHAP-ranked genes of at least one model. Of the 64 genes in the final panel, 14 satisfied all three criteria and were nominated as candidate biomarkers (Table 4.5). All 14 were both significantly differentially expressed and retained across all five iterations. They differed in which models ranked them highly. *ABCG4* was the only candidate appearing in every model's top ten. *GDA*, *TAP2*, and *HOXC10* reached the top ten of the support vector machine only. Because the SVM's Kernel SHAP estimates did not converge (Section 4.2), its rankings are the least reliable of the three, so the importance evidence for these three candidates is weaker than for genes ranked highly by the tree-based models. Their directional

effects were nonetheless consistent with their differential expression ($|r| \geq 0.97$), providing some support.

Table 4.5: The 14 candidate biomarkers meeting all three convergence criteria. Each was significantly differentially expressed (FDR-adjusted $p < 0.05$, $|\log_2FC| > 1$) and retained by the elastic-net filter in all five cross-validation outer iterations. SHAP ranks are each gene's position by mean absolute SHAP value within each model (— = not ranked / zero importance; for XGBoost, only 7 genes had non-zero importance). The final column lists the models placing the gene in their top ten (RF = random forest, XGB = XGBoost, SVM = support vector machine).

Gene	Iterations	Direction	Log ₂ FC	Adjusted p-value	RF rank	XGB rank	SVM rank	Top 10 (models)
ABCG4	5	Up	1.65	6.72×10^{-82}	3	1	9	RF, XGB, SVM
S100A8	5	Up	8.38	2.37×10^{-102}	2	—	44	RF
S100A12	5	Up	5.83	9.52×10^{-112}	8	—	42	RF
IL1F9	5	Up	5.20	2.28×10^{-107}	7	—	45	RF
AKR1B10	5	Up	5.19	4.03×10^{-118}	17	2	48	XGB
FABP5	5	Up	3.41	9.12×10^{-95}	5	—	46	RF
GDA	5	Up	2.90	1.42×10^{-78}	45	—	4	SVM
AKR1B15	5	Up	2.89	7.46×10^{-98}	14	3	31	XGB
VNN3	5	Up	2.23	3.03×10^{-87}	1	—	26	RF
COBL	5	Down	-1.53	6.31×10^{-81}	13	5	47	XGB
TPBG	5	Up	1.47	3.00×10^{-86}	4	—	15	RF
CDK5R1	5	Up	1.36	3.65×10^{-93}	16	6	40	XGB
TAP2	5	Up	1.13	5.02×10^{-66}	48	—	6	SVM
HOXC10	5	Down	-1.05	1.62×10^{-49}	56	—	2	SVM

To place these candidates in context, DISGENET (MedBioinformatics Solutions, 2026a) was queried for each gene as external annotation rather than as a selection criterion (Table 4.6). Eight of the 14 candidates were already associated with psoriasis in DISGENET, including *IL1F9*, *S100A8*, *TAP2*, *FABP5* and *VNN3*. The remaining six, *ABCG4*, *AKR1B15*, *COBL*, *TPBG*, *CDK5R1* and *HOXC10* had no recorded psoriasis association and are therefore potentially novel candidates arising from the convergence analysis. Notably, *ABCG4*, the only gene in all three models' top ten, was among these. Its strongest recorded disease associations were with prostate cancer and Alzheimer's disease rather than any skin condition. The complete DISGENET export files for all 14 candidate biomarkers are provided as supplementary material in the accompanying Google Drive.

Table 4.6: DISGENET gene–disease associations for the 14 candidate biomarkers, restricted to entries typed as diseases (phenotypes excluded). The psoriasis GDA score is the gene–disease association score for psoriasis (0–1; — = no recorded psoriasis association in DISGENET, i.e. potentially novel), and the psoriasis PMIDs column gives the number of supporting publications for that association. DSI is the gene-level Disease Specificity Index (higher = associated with fewer diseases). The final column lists each gene's three highest-scoring associated diseases, with the GDA score and number of supporting publications (PMIDs) for each; ties in GDA score are broken by descending PMID count. Where the same disease appeared twice under the same gene, these were merged and the highest PMID count is reported. Two genes are recorded in DISGENET under their current HGNC symbols: IL1F9 as IL36G and VNN3 as VNN3P.

Gene	Psoriasis GDA Score	Psoriasis PMIDs	DSI	Top three associated diseases (GDA score, PMIDs)
ABCG4	—	—	0.71	Prostate cancer (0.55, 8 PMIDs); Alzheimer's disease (0.25, 2 PMIDs); Pulmonary alveolar proteinosis (0.15, 1 PMID)
S100A8	0.35	27	0.41	Acute myeloid leukaemia (0.75, 14 PMIDs); Atopic dermatitis (0.75, 8 PMIDs); Myocarditis (0.65, 3 PMIDs)
S100A12	0.25	8	0.47	Inflammatory bowel disease (0.45, 22 PMIDs); Migraine disorders (0.45, 10 PMIDs); Crohn's disease (0.40, 12 PMIDs)
IL1F9	0.35	32	0.56	Psoriasis (0.35, 32 PMIDs); Atopic dermatitis (0.35, 7 PMIDs); Interleukin-36 receptor antagonist deficiency (0.25, 6 PMIDs)
AKR1B10	0.30	2	0.54	Liver cancer (0.85, 55 PMIDs); Endometrial cancer (0.40, 1 PMID); Colorectal cancer (0.40, 1 PMID)
FABP5	0.40	22	0.51	Liver cancer (0.70, 25 PMIDs); Schizophrenia (0.55, 1 PMID); Prostate cancer (0.40, 23 PMIDs)
GDA	0.10	1	0.70	Carcinoma (0.40, 1 PMID); Epithelioma (0.40, 1 PMID); Drug-induced liver injury (0.40, 1 PMID)
AKR1B15	—	—	0.83	Liver cancer (0.10, 1 PMID); Acute kidney injury (0.10, 1 PMID); Colon cancer (0.10, 1 PMID)
VNN3	0.40	1	0.76	Psoriasis (0.40, 1 PMID); Malaria (0.15, 1 PMID); ST-elevation myocardial infarction (0.10, 1 PMID)
COBL	—	—	0.72	Type 2 diabetes mellitus (0.15, 2 PMIDs); Mental disorders (0.10, 1 PMID); Bipolar disorder (0.10, 1 PMID)
TPBG	—	—	0.66	Prostate cancer (0.50, 1 PMID); Carcinoma (0.25, 9 PMIDs); Colorectal cancer (0.20, 6 PMIDs)
CDK5R1	—	—	0.65	Alzheimer's disease (0.70, 7 PMIDs); Schizophrenia (0.55, 1 PMID); Encephalopathies (0.40, 1 PMID)
TAP2	0.80	6	0.45	Type 1 diabetes mellitus (0.90, 16 PMIDs); Multiple sclerosis (0.90, 8 PMIDs); Alzheimer's disease (0.80, 18 PMIDs)
HOXC10	—	—	0.63	Gastric cancer (0.20, 9 PMIDs); Colorectal cancer (0.20, 6 PMIDs); Breast cancer (0.20, 5 PMIDs)

In summary, differential expression identified 1,131 significantly dysregulated genes, including the most strongly upregulated and downregulated genes reported in Tables 4.1 and 4.2 (RQ1), all three classifiers separated the tissue types without error while relying on mostly different gene sets (RQ2), and combining differential expression, selection stability and model-based SHAP importance yielded 14 candidate biomarkers, 8 with recorded psoriasis associations in DISGENET and 6 potentially novel (RQ3). These findings are interpreted in relation to the literature in Chapter 5.

Chapter 5: Discussion

The previous chapter presented the results of the study. This chapter interprets those results, assessing what they reveal about the study’s research questions and how they relate to the existing psoriasis literature. It is organised around the three research questions: differential expression (Section 5.1), classification and cross-model agreement (Section 5.2), and the candidate biomarkers and their disease associations (Section 5.3).

5.1 Differential Expression Analysis

Differential expression analysis conducted in this study recovered the established gene expression signature of psoriasis, giving confidence that the machine learning models, which ran on the same data, were working from a reliable starting point. Of the 21,099 tested genes, 1,131 met both significance thresholds (FDR-adjusted $p < 0.05$ and $|\log_2FC| > 1$; 489 upregulated and 642 downregulated). This scale of dysregulation is consistent with the substantial number of genes that are reported as differentially expressed in psoriatic skin compared to normal skin (Li et al., 2014). More persuasive than the count is which genes show the largest changes. The most upregulated genes in this study were *DEFB4A*, *PI3*, several *S100* family members and the *SERPINE* genes, and these antimicrobial and inflammatory genes are well-documented as being elevated in psoriatic skin (Furie et al., 2020; He et al., 2025; Rioux et al., 2020; Xiao et al., 2024; Zhao et al., 2024a). That the largest magnitude changes fall on precisely these genes, all with FDR-adjusted p -values below 10^{-89} , indicates that the differential expression step reflects established biology rather than dataset-specific noise. Due to the coordinated dysregulation of numerous genes in psoriatic skin, Chapter 2 did not discuss *PI3*, as it focused on the most frequently cited genes rather than an exhaustive set. *PI3* was also among the most strongly upregulated genes in this study, consistent with reports that its protein

(peptidase inhibitor 3) recurs among the most upregulated genes in the literature (Rioux et al., 2020) and is linked to the abnormal build-up of skin cells characteristic of the disease (Deng et al., 2023). Its unprompted discovery is a further sign that the analysis reflects genuine disease biology.

The downregulated genes derived from the literature provide a second, more stringent check. Because the dysregulation magnitudes were far lower among the downregulated genes than the upregulated ones (Section 4.1), identifying these subtler changes was a harder task. *WIF1* and *BTC*, both consistently reported as downregulated in psoriatic skin (Gudjonsson et al., 2010a, 2010b; Ma et al., 2024a; Peng et al., 2022; Suárez-Fariñas et al., 2012; Zhang et al., 2025), where their reduced expression is linked to plaque development (Liu et al., 2023; Peng et al., 2022), were significantly downregulated in this study. However, neither ranked among the ten largest-magnitude changes (*WIF1* was ranked 22nd and *BTC* was 16th. Available on the *limma_sig_genes_downregulated.csv* file on the Google Drive). As expected, their reduction was reproducible in direction but modest in magnitude. This is consistent with Section 2.2.1, where fold changes varied across studies due to different profiling technologies and cohorts. Identifying these genes as being downregulated is itself evidence that the differential expression analysis was accurate. The downregulated set, led by *THRSP*, *CCL27* and *KRT77*, was far more modest in magnitude than the upregulated set, extending only to a log₂FC of roughly -3.8. This indicates that dysregulation is asymmetric. All three genes are established as downregulated in psoriatic skin (Gudjonsson et al., 2010a; Rioux et al., 2020; Suárez-Fariñas et al., 2012), consistent with the validation seen for *WIF1* and *BTC*.

Two clarifications frame the analysis that follows. First, the asymmetry above means that the gene expression differences between psoriatic and normal skin are dominated by the strongly upregulated genes. More importantly, differential expression establishes only that a gene differs between the two groups; it does not establish that a gene is predictively important (Section 2.2.1). That distinction motivates the machine learning analysis, which asks not which genes differ, but which genes the machine learning algorithms rely on, thus narrowing the differentially expressed set to those the algorithms actually depend on for classification.

5.2 Classification Performance and Cross-Model Agreement (RQ2)

All three classifiers reached 1.00 on every performance metric, under both nested cross-validation (with zero variance across the five outer folds) and on the held-out test set. This uniform, ceiling-level performance is expected rather than notable. Within-cohort classification reaches or approaches the ceiling across the literature, including two prior studies that applied machine learning to this dataset (GSE54456) (He et al., 2024; Tang et al., 2025). This strong within-cohort performance is due to the classification signal being strong and spread across many correlated genes, as established in Section 2.2.2. The exploratory data analysis had already foreshadowed this, as the first principal component alone separated the two tissue types (Section 3.3.5), so the data were close to linearly separable before any model was trained, making perfect classification performance expected. Critically, this strong performance, obtained under leakage-controlled nested cross-validation, and consistent with prior studies conducted on this dataset, establishes only that each model classifies reliably; it says nothing about which genes are important for classification. The perfect performance metrics are therefore best understood as a prerequisite for the interpretation that follows, rather than the main result of this machine learning analysis.

The informative comparison is therefore not how well the models classify but which genes they rely upon, and here the three models disagreed substantially. *ABCG4* was the only gene ranked within the top ten of all three (XGBoost 1st, random forest 3rd, SVM 9th). Every other top-ranked gene was specific to a single model. *VNN3*, for example, was the random forest's most important gene yet ranked 26th under the SVM and was assigned no importance by XGBoost, while *TAP2*, *GDA*, and *HOXC10* were only in the SVM model's top ten.

This disagreement on feature importance between the models is not a shortcoming of the analysis but one of its central findings. As established in Section 2.2.2, single-model importance rankings are unreliable for two independent reasons: correlation between genes distorts the importance any one model assigns (Tološi & Lengauer, 2011), and different algorithms favour different features in any case (Saarela & Jauhiainen, 2021). An importance ranking from one algorithm may therefore reflect that algorithm's bias rather than genuine predictive importance, which is precisely why single-model interpretation, as seen in He et al. (2024) and Liu et al. (2026), is insufficient on its own. Seen in this light, the disagreement

between this study's three models is informative in its own right. Its breadth is consistent with many correlated genes carrying the same signal (Section 2.2.2), so that different algorithms draw on different members of the same correlated groups; the rare instance of agreement, by contrast, isolates the gene least susceptible to any single algorithm's bias.

Two aspects of these results warrant caution. First, XGBoost assigned non-zero importance to only 7 of the 64 genes, so it could not agree with the other models on the remaining 57 genes. The limited cross-model agreement therefore partly reflects this sparsity rather than genuine disagreement. Second, the SVM's Kernel SHAP estimates did not converge, so its importance magnitudes and ranks are the least reliable of the three. Both points reinforce the principle that no single algorithm's feature importance is fully reliable, making comparison across all three models, combined with the other lines of evidence, the more appropriate basis for nominating biomarker candidates.

5.3 Candidate Biomarkers and Disease Associations (RQ3)

Combining differential expression, feature selection stability, and classifier feature importance yielded 14 convergent candidate biomarkers, each satisfying all three criteria. Each of the 14 genes differed significantly in expression between psoriatic and normal skin, was retained by the elastic net filter in all five outer iterations of the nested cross-validation, so none depended on a single train-test partition and was ranked among the top ten by mean absolute SHAP value in at least one algorithm. Notably, the machine learning models diverged in the importance they attributed to these stably selected genes (Section 5.2), yet all 14 cleared this shared threshold. They are divided into two groups based on whether an existing psoriasis association is recorded in DISGENET (MedBioinformatics Solutions, 2026a). The genes that have established associations to psoriasis biology validate the analysis, while those that do not are the study's most novel finding. A recurring property across both groups is pleiotropy: association with diseases beyond psoriasis.

Genes with established psoriasis associations:

The first group consists of candidates that are already linked to psoriasis, whose identification confirms that this dissertation's pipeline finds genes known to be involved in the disease. *FABP5* is a great example of this. It has been found to be upregulated in psoriatic skin, and its protein has been functionally implicated in keratinocyte hyperproliferation and inflammation (Mieczkowski et al., 2025; Warren et al., 2025). It is one of the genes that is the most consistently selected by machine learning models across prior psoriasis studies (Le et al.,

2021; Li et al., 2026; Ma et al., 2024b), so its emergence here as a candidate biomarker confirms that this dissertation’s analysis pipeline reproduces both known disease biology and the findings of other machine learning studies. *IL1F9*, recorded in DISGENET under its current symbol (*IL36G*), is a well-established psoriasis inflammatory gene. Its association with psoriasis is supported by 32 publications (Section 4.3), far more than any other candidate. Furthermore, *S100A8* is consistently upregulated in psoriatic skin, where its protein drives inflammatory signalling in keratinocytes and immune cells (He et al., 2025; Zhang et al., 2025; Zhao et al., 2024a) and was the second most important gene in the random forest model (Section 4.2). *S100A12*, another *S100*-family gene, is likewise upregulated in psoriatic skin (Roberson et al., 2012) and has surfaced at the feature selection stage of prior psoriasis machine learning studies (Hao et al., 2024; Zhao et al., 2024b). That these dysregulated genes are also among the most important to the machine learning analysis matters because agreement between two independent analyses (differential expression and classification) is stronger evidence of real disease biology than either method can provide alone. The evidence is not uniform across this group, however. Two candidates are only nominally associated with psoriasis: *VNN3* (recorded in DISGENET under its current symbol *VNN3P*) carries a score of 0.40, and *GDA* (guanine deaminase) a score of 0.10. Each rests on a single publication, so both remain effectively understudied in this context, notably *VNN3*, despite being the random forest’s most important gene.

Several genes in this group are pleiotropic, their DISGENET associations pointing well beyond the skin. *TAP2* carries the strongest psoriasis association of any candidate (a high GDA score of 0.80) yet is equally linked to type 1 diabetes and multiple sclerosis (both 0.90), autoimmune conditions that share the antigen-processing pathway *TAP2* belongs to, and it is also associated with Alzheimer’s disease (0.80). *AKR1B10*, which has itself appeared among selected features in prior psoriasis machine learning work (Hao et al., 2024), along with *FABP5*, are both associated most strongly with liver cancer (0.85 and 0.70) rather than any skin condition; *FABP5* is further linked to schizophrenia (0.55). *S100A12* associates most strongly with inflammatory bowel disease (0.45). These cross-disease associations do not diminish the genes’ relevance to psoriasis, but show them broadly implicated across inflammatory, neoplastic, and psychiatric conditions. This is the same cross-disease pattern established in Section 2.2.1.

Genes with no recorded psoriasis association:

The second group, the most novel finding of this study, is comprised of six candidates that have no association with psoriasis in DISGENET: *ABCG4*, *AKR1B15*, *COBL*, *TPBG*, *CDK5R1* and *HOXC10*. Here, ‘no recorded association’ refers specifically to DISGENET’s curated and text-mined records. It marks the absence of an established, catalogued link rather than proof that no association exists anywhere in the literature. These are genes not yet established in psoriasis and therefore warrant further investigation rather than confirmed associations. *ABCG4* is the most striking. It was the only gene ranked in the top ten of all three models yet has no recorded psoriasis association. Its strongest links are to prostate cancer (GDA of 0.55) and Alzheimer’s disease (small association at 0.25), an instance of pleiotropy seen above, but here in a gene with no psoriasis association. Beyond its cross-model ranking, *ABCG4*’s predictive behaviour was also consistent in direction. Across all three models, higher expression drove predictions towards psoriasis (Section 4.2), matching with its upregulation in the differential expression analysis. For a gene with no recorded psoriasis association, this agreement between expression change and predictive direction is the strongest evidence available from this study that its predictive signal is real rather than a chance pattern. It does not, however, establish *ABCG4* as the causal driver over other correlated genes carrying the same signal, and its role in psoriasis remains to be tested directly. The remaining candidates carry their own non-psoriatic associations: *CDK5R1* links most strongly to Alzheimer’s disease (0.70), *HOXC10* to gastric cancer (0.20) and *TPBG* to prostatic neoplasms (0.50). As shown in Section 2.2.2, prior psoriasis machine learning studies typically restrict their input to a pre-defined gene category before training (He et al., 2024; Liu et al., 2026; Tang et al., 2025). Identifying informative, understudied genes that these restricted category pipelines could not, whether because they fell outside the chosen categories, were not retained among the final selected features, or were overlooked by a single model, is the direct payoff of this study’s genome-wide search and its use of feature importance across multiple models.

However, a caveat applies to all 14 candidates. These genes are dysregulated in psoriatic skin and predictive of the disease, but neither differential expression nor prediction establishes a role in psoriasis pathogenesis. Expression-based analysis cannot distinguish a gene that causes psoriasis from one that merely changes as a consequence of the disease, or one that simply correlates with the true drivers. This caveat does not weaken the established genes, whose involvement is supported by prior functional studies. However, it is the novel candidates that most warrant laboratory investigation, as they lack prior evidence. Among them, *ABCG4*

is the strongest lead. *GDA*, *TAP2*, and *HOXC10* carry the greatest uncertainty, as each was ranked highly by the support vector machine alone, whose non-convergence makes its rankings the least reliable.

Taken together, the three research questions form a single argument. Differential expression validated the pipeline by recovering the established psoriasis transcriptome (RQ1). Classification was perfect across all models and so classification performance could not distinguish them, but their feature-importance rankings could, and although the models disagreed widely on importance, a subset of genes reached the top ten of at least one model, and *ABCG4* alone was important to all three (RQ2). Combining these techniques (differential expression, stability across all five cross-validation outer iterations, and a top ten importance ranking in at least one model) defined 14 biomarker candidates, which DISGENET annotation then sorted into established psoriasis genes and genes with no recorded psoriasis link, and revealed several associations extending across other inflammatory, neoplastic, and neurological conditions (RQ3). By searching genome-wide rather than within pre-restricted gene categories, and by drawing importance from several classification models rather than one, this study identified candidates that prior single-model, category-restricted studies could not, most notably genes such as *ABCG4*, which have no recorded psoriasis association. These novel candidates warrant functional investigation rather than being treated as confirmed contributors to psoriasis. Limitations and further work are considered in Chapter 6.

Chapter 6: Conclusion

This dissertation set out to identify robust candidate biomarkers of psoriasis through a genome-wide search requiring convergence across differential expression analysis, cross-validation selection stability, and machine learning feature importance, and to examine the wider disease associations of those genes. This chapter returns to that aim and to the three research questions, summarises the study's contribution and implications, offers a short ethical reflection, and sets out the limitations and the future work they motivate.

The study addressed its three research questions and, in doing so, met each objective. Differential expression analysis recovered the established psoriasis transcriptome, identifying 1,131 significantly dysregulated genes led by the well-documented antimicrobial and inflammatory genes *DEFB4A*, *PI3*, and the *S100* and *SERPINB* families (RQ1). The three classifiers each separated psoriatic from non-psoriatic skin without error under leakage-controlled nested cross-validation, but their feature importance rankings diverged sharply, with

ABCG4 the only gene important to all three (RQ2). Combining the three lines of evidence yielded 14 candidate biomarkers, which DISGENET annotation then sorted into eight genes with established links to psoriasis and six with none (RQ3). The study therefore achieved its aim of identifying robust candidate biomarkers by converging three distinct criteria and mapping their broader disease associations.

The central contribution of this study is methodological as much as biological. By searching genome-wide rather than a pre-defined gene category, and by comparing feature importance across three independent algorithms rather than interpreting a single model, this dissertation surfaced candidates that earlier studies could not identify, because those studies restricted the gene pool to pre-defined categories and applied post-hoc interpretation to a single model at most. The six genes with no recorded association with psoriasis, most notably *ABCG4*, are this dissertation's most novel finding. The recurring pleiotropy of the candidates, with associations spanning inflammatory, neoplastic, and neurological conditions, also situates psoriasis within a wider network of related diseases. Practically, these candidates offer a prioritised, evidence-weighted shortlist for laboratory follow-up, a more efficient starting point than the full differentially expressed set. Should these candidates be validated, they could support both a clearer understanding of psoriasis mechanisms and the search for new therapeutic targets.

This study also involved ethical considerations. It used only anonymised, publicly available secondary data with no personal identifiers, in line with University of Sheffield policy, so no individual could be identified or harmed. However, this anonymity has a scientific cost. Without demographic information on the cohort, it cannot be established whether the candidate biomarkers generalise across ethnicity, age, and sex, and biomarkers identified from one cohort may not apply to external cohorts. Additionally, the novel biomarkers identified in this study have been appropriately reported, as premature claims about their clinical value could mislead patients, clinicians, and future research.

Three limitations of this study must be considered. Each one also indicates a direction for future work. First, and most importantly, the analysis utilised a single cohort (GSE54456), with no external validation dataset. As the literature review established, strong within-cohort classification performance is nearly guaranteed because psoriatic and normal skin differ in the expression of thousands of genes, so the two classes are easily separated within any single dataset. Relying on one cohort therefore offers limited evidence that the findings would generalise. The 14 candidates are therefore well supported within this dataset but not yet externally validated. The clear priority for future work is to replicate the convergence pipeline

across multiple independent cohorts and, ideally, different profiling platforms (such as microarrays and RNA sequencing), retaining only candidates that hold across datasets. Second, the support vector machine's Kernel SHAP estimates did not converge across the 34 test samples, because approximating Shapley values over 64 genes is computationally expensive, so its exact ranks and magnitudes were treated with caution. Because the SVM is linear, however, its SHAP values could alternatively be computed exactly from the decision function (Linear SHAP), which future work could adopt to avoid the non-convergence encountered in this study.

Third, the analysis is fundamentally associative. Expression-based evidence cannot distinguish a gene that drives psoriasis from one that changes as a consequence of it, or one that merely correlates with the true drivers. The novel candidates lack the functional support behind the established psoriasis genes, so laboratory validation is needed in order to determine whether they are causal or merely associative with the disease.

In closing, this dissertation demonstrates that searching the full transcriptome, requiring agreement across several methods, and annotating disease associations can move beyond confirming that psoriasis is classifiable towards helping identify which genes matter in psoriasis and why. This study's main contribution is the pipeline it demonstrates: a disciplined, leakage-controlled route from thousands of dysregulated genes to a small set of prioritised candidates that warrant further laboratory testing.

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Appendix A: Genes Selected by Elastic Net in Each Cross-Validation Outer Iteration

Genes selected by the elastic-net filter ($\alpha = 0.8$, λ at the one-standard-error rule) in each of the five outer cross-validation iterations, ordered alphabetically within each iteration. The final row reports the total number of genes selected in each iteration. Because a fixed seed and shared folds were used, these selections were identical across all three classification models.

Iteration 1	Iteration 2	Iteration 3	Iteration 4	Iteration 5
ABCG4	ABCG4	ABCG4	ABCG4	ABCG4
ADAP2	ADAMDEC1	ADAMDEC1	ADAMDEC1	ADAMDEC1
AKR1B10	ADAP2	ADAP2	AKR1B10	AKR1B10
AKR1B15	AKR1B10	AKR1B10	AKR1B15	AKR1B15
ANKFN1	AKR1B15	AKR1B15	BTC	ANKFN1
AS3MT	ANKFN1	ARSF	C10orf99	AQP9
BTC	BDNFOS	BIRC5	C6orf26	ARSF
C10orf99	BTC	BTC	C9orf122	BTC
C6orf26	C10orf99	C10orf99	CCR7	C10orf99
C9orf122	C6orf26	C6orf26	CDK5R1	C5orf46
CACNA2D2	C9orf122	CACNA2D2	CHAD	C6orf26
CDK5R1	CACNA2D2	CCNB2	CHI3L2	CCL22
CHAD	CCBP2	CCR7	CLDN8	CCR7
CHI3L2	CCR7	CDK5R1	CLEC7A	CDK5R1
CLEC7A	CDK5R1	CHI3L2	COBL	CHI3L2
COBL	CHI3L2	CHRNA9	CXCL9	CLDN1
CXCR6	CLDN8	CLDN8	CXCR6	CLEC7A
CYP2W1	CLEC7A	COBL	CYP2W1	COBL
DEFB4A	COBL	CXCL9	DEFB4A	CXCR4
EPHB1	CXCL9	CXCR6	EPHB1	CXCR6
F3	CXCR6	CYP2W1	F3	DEFB4A
FABP5	CYP2J2	DEFB4A	FABP5	F3
FAM110C	CYP2W1	EPHB1	GBP1	FABP5
FUT3	DEFB4A	EPR1	GDA	FAM110C
GDA	EPHB1	F3	GJB2	FOXP3
GJB2	EPR1	FABP5	GRHL3	FUT3
GRHL3	F3	FOXM1	HOXC10	GBP5
GZMB	FABP5	FUT3	HPSE	GDA
HOXC10	GBP1	GDA	IL1F9	GJB2
HPSE	GDA	GJB2	INA	GRHL3
IFI27	GJB2	GPR1	KYNU	HOXC10
IL1F9	GRHL3	HOXC10	LCE3A	IL1F9
INA	HOXC10	HPSE	LOC731779	INA
KYNU	HPSE	IL1F9	PHYHIP	KRT77
LCE3A	IL1F9	INA	PI3	KYNU
MAP6	INA	KIF20A	PLA2G4D	LCE3A
MSMB	IRF1	KIF2C	PLBD1	LTF
PHYHD1	KYNU	KYNU	PLLP	MPZL2
PHYHIP	LCE3A	LCE3A	RAB3B	MSMB
PI3	LOC400759	LCE3D	RRM2	PI3
PLA2G4D	LOC731779	LCE3E	S100A12	RAB3B
PLBD1	MSMB	LOC731779	S100A7	RGMB
PLLP	PHYHIP	MSMB	S100A7A	RGS1
RAB3B	PI3	PHYHIP	S100A8	RORC
RGMB	PLA2G4D	PI3	S100A9	S100A12
S100A12	PLBD1	PLA2G4D	SERPINB13	S100A7
S100A7A	PLLP	PLBD1	SERPINB3	S100A7A
S100A8	RAB3B	PLLP	SERPINB4	S100A8

Iteration 1	Iteration 2	Iteration 3	Iteration 4	Iteration 5
S100A9	RGMB	RRM2	SOX7	S100A9
SERPINB13	S100A12	S100A12	SPRR2A	SERPINB4
SLC7A1	S100A7	S100A7A	SPRR2B	SLC7A1
SOX7	S100A7A	S100A8	TAP1	SNTB1
SPRR2A	S100A8	S100A9	TAP2	SPRR2A
TAP1	S100A9	SERPINB3	TCN1	SPRR2B
TAP2	SLC7A1	SERPINB4	TMEM61	SPRR2D
TMEM61	SPRR2A	SOX7	TPBG	TAP2
TPBG	SPRR2B	SPRR2A	ULBP2	TPBG
TRIM15	STAR	SPRR2B	VNN3	TRIM10
VNN3	TAP1	SPRR2G	WDR72	TTC39A
WDR72	TAP2	TAP2	WNT5A	VNN3
WNT5A	TMEM61	TLE1		WNT5A
	TPBG	TMEM61		ZC3H12A
	TREX2	TPBG		
	TRIM15	TPX2		
	VNN3	TREX2		
	WNT5A	TRIM15		
		TTC39A		
		UGT3A2		
		VNN3		
		WDR72		
		WNT5A		
61	66	71	60	62

Appendix B: The Stable Core of 30 Genes

Selected Across all Five Cross-Validation Iterations

The stable core of 30 genes selected by the elastic-net filter in all five outer cross-validation iterations, ordered alphabetically. As in Appendix A, this selection was identical across all three classification models.

Gene		
ABCG4	DEFB4A	PI3
AKR1B10	FABP5	SPRR2A
AKR1B15	F3	S100A7A
BTC	GDA	S100A8
CDK5R1	GJB2	S100A9
CHI3L2	HOXC10	S100A12
COBL	IL1F9	TAP2
CXCR6	INA	TPBG
C6orf26	KYNU	VNN3
C10orf99	LCE3A	WNT5A

Appendix C: The Final 64-gene Panel Selected on the Full Training Set

The final 64-gene panel selected by the elastic net filter ($\alpha = 0.8$) on the full training set, and identical across all three classification models. Ordered alphabetically.

Gene			
ABCG4	CLDN8	IL1F9	S100A7A
ADAMDEC1	CLEC7A	INA	S100A8
ADAP2	COBL	IRF1	S100A9
AKR1B10	CXCL9	KYNU	SERPINB4
AKR1B15	CXCR6	LCE3A	SLC7A1
ANKFN1	CYP2W1	LOC731779	SOX7
ARSF	DEFB4A	LTF	SPRR2A
BTC	EPHB1	MSMB	SPRR2B
C10orf99	F3	PHYHIP	TAP1
C6orf26	FABP5	PI3	TAP2
C9orf122	FUT3	PLA2G4D	TMEM61
CACNA2D2	GDA	PLBD1	TPBG
CCR7	GJB2	PLLP	TRIM15
CDK5R1	GRHL3	RAB3B	VNN3
CHAD	HOXC10	RRM2	WDR72
CHI3L2	HPSE	S100A12	WNT5A

Appendix D: Ethics Approval Letter

Ethics approval letter from the University of Sheffield Research Ethics Committee (application reference 074624), confirming ethical approval for this project.



Downloaded: 17/08/2026
Approved: 08/06/2026

Cailey Coghlan
Registration number: 250125144
School of Information, Journalism and Communication
Programme: MSc Data Science

Dear Cailey

PROJECT TITLE: Identifying Psoriasis Gene Drivers Through Differential Expression Analysis, Machine Learning, and Post Hoc Interpretability
APPLICATION: Reference Number 074624

This letter confirms that you have signed a University Research Ethics Committee-approved self-declaration to confirm that your research will involve only existing research, clinical or other data that has been robustly anonymised. You have judged it to be unlikely that this project would cause offence to those who originally provided the data, should they become aware of it.

As such, on behalf of the University Research Ethics Committee, I can confirm that your project can go ahead on the basis of this self-declaration.

If during the course of the project you need to [deviate significantly from the above-approved documentation](#) please inform me since full ethical review may be required.

Yours sincerely

Ijc Ethics
Departmental Ethics Administrator

Appendix E: Convergence Results for all 64 genes

Full convergence results for all 64 genes. The 14 genes meeting all three convergence criteria (the candidate biomarkers reported in Table 4.5) are listed first, followed by the remaining genes ordered by the number of criteria met and then by absolute $\log_2$ fold change. The three criteria were: significant differential expression (FDR-adjusted $p < 0.05$ and $|\log_2FC| > 1$); retention by the elastic-net filter in all five outer cross-validation iterations; and placement within the top ten SHAP-ranked genes of at least one model. Iterations gives the number of outer cross-validation iterations in which the gene was retained. SHAP ranks are each gene's position by mean absolute SHAP value within each model (— = not ranked or zero importance; for XGBoost, only seven genes received non-zero importance). Criteria met is the number of the three criteria satisfied.

Gene	Direction	Log ₂ FC	Adjusted P-Value	Iterations	RF rank	XGB rank	SVM rank	Criteria met
ABCG4	Up	1.65	6.72×10^{-82}	5	3	1	9	3
S100A8	Up	8.38	2.37×10^{-102}	5	2	—	44	3
S100A12	Up	5.83	9.52×10^{-112}	5	8	—	42	3
IL1F9	Up	5.20	2.28×10^{-107}	5	7	—	45	3
AKR1B10	Up	5.19	4.03×10^{-118}	5	17	2	48	3
FABP5	Up	3.41	9.12×10^{-95}	5	5	—	46	3
GDA	Up	2.90	1.42×10^{-78}	5	45	—	4	3
AKR1B15	Up	2.89	7.46×10^{-98}	5	14	3	31	3
VNN3	Up	2.23	3.03×10^{-87}	5	1	—	26	3
COBL	Down	-1.53	6.31×10^{-81}	5	13	5	47	3
TPBG	Up	1.47	3.00×10^{-86}	5	4	—	15	3
CDK5R1	Up	1.36	3.65×10^{-93}	5	16	6	40	3
TAP2	Up	1.13	5.02×10^{-66}	5	48	—	6	3
HOXC10	Down	-1.05	1.62×10^{-49}	5	56	—	2	3
DEFB4A	Up	10.57	2.81×10^{-115}	5	23	—	22	2
PI3	Up	9.41	4.81×10^{-114}	5	29	—	58	2
S100A9	Up	8.91	3.82×10^{-105}	5	21	—	52	2
S100A7A	Up	8.12	6.88×10^{-125}	5	25	—	35	2
LCE3A	Up	7.55	1.94×10^{-109}	5	19	—	53	2
SPRR2A	Up	6.00	6.85×10^{-105}	5	42	—	39	2
C10orf99	Up	5.16	2.68×10^{-103}	5	11	—	41	2
GJB2	Up	4.12	4.59×10^{-100}	5	20	—	57	2
CHI3L2	Up	3.90	2.70×10^{-72}	5	55	—	19	2
KYNU	Up	3.46	1.05×10^{-93}	5	18	—	37	2
BTC	Down	-2.94	7.12×10^{-95}	5	27	—	28	2
C6orf26	Down	-2.46	5.23×10^{-79}	5	22	—	13	2
PHYHIP	Down	-2.35	2.23×10^{-61}	4	38	—	3	2
ADAMDEC1	Up	2.26	2.64×10^{-60}	4	47	7	64	2
F3	Down	-2.19	2.74×10^{-75}	5	31	—	12	2
FUT3	Up	2.04	8.64×10^{-81}	3	9	—	59	2
CYP2W1	Down	-2.00	1.28×10^{-76}	4	10	—	11	2
WNT5A	Up	1.76	3.83×10^{-72}	5	26	—	21	2
ADAP2	Up	1.70	1.51×10^{-84}	3	36	4	43	2
INA	Up	1.50	3.32×10^{-71}	5	37	—	27	2
CXCR6	Up	1.39	4.25×10^{-58}	5	49	—	50	2
SLC7A1	Up	1.34	5.22×10^{-74}	3	6	—	36	2
SOX7	Up	1.27	1.75×10^{-59}	3	46	—	10	2

Gene	Direction	Log ₂ FC	Adjusted P-Value	Iterations	RF rank	XGB rank	SVM rank	Criteria met
ANKFN1	Down	-1.06	2.27×10^{-66}	3	35	—	8	2
SERPINB4	Up	8.31	5.05×10^{-108}	3	12	—	55	1
SPRR2B	Up	5.56	1.39×10^{-98}	4	39	—	38	1
LTF	Up	4.82	2.37×10^{-58}	1	—	—	63	1
PLA2G4D	Up	4.07	2.93×10^{-94}	4	24	—	56	1
MSMB	Down	-3.11	3.91×10^{-58}	4	—	—	61	1
HPSE	Up	2.77	6.27×10^{-93}	4	28	—	49	1
ARSF	Up	2.73	1.07×10^{-71}	2	44	—	60	1
CXCL9	Up	2.24	5.49×10^{-36}	3	—	—	32	1
RRM2	Up	2.07	4.53×10^{-75}	2	40	—	62	1
PLL	Down	-1.90	1.09×10^{-63}	4	33	—	17	1
CLEC7A	Up	1.83	3.64×10^{-65}	4	51	—	29	1
PLBD1	Up	1.77	3.70×10^{-80}	4	43	—	20	1
CLDN8	Down	-1.75	3.95×10^{-59}	3	50	—	23	1
CHAD	Down	-1.74	1.45×10^{-73}	2	32	—	33	1
CCR7	Up	1.68	1.22×10^{-54}	4	52	—	54	1
GRHL3	Up	1.50	9.95×10^{-64}	4	41	—	51	1
RAB3B	Down	-1.18	1.12×10^{-65}	4	34	—	14	1
TRIM15	Up	1.16	4.63×10^{-60}	3	15	—	34	1
IRF1	Up	1.03	8.15×10^{-40}	1	—	—	16	1
WDR72	Down	-1.01	1.04×10^{-45}	3	—	—	18	1
TAP1	Up	0.94	1.70×10^{-44}	3	—	—	1	1
TMEM61	Down	-0.84	1.66×10^{-40}	4	—	—	7	1
LOC731779	Down	-0.42	1.08×10^{-36}	3	—	—	5	1
CACNA2D2	Down	-0.95	1.73×10^{-68}	3	30	—	24	0
EPHB1	Down	-0.92	5.54×10^{-48}	4	54	—	30	0
C9orf122	Down	-0.59	1.03×10^{-51}	3	53	—	25	0

Appendix F: Research Diary and Reflection

This appendix records the four supervision meetings held with Dr. Bath during the dissertation period and reflects on five key instances of feedback, the actions taken in response, and the skills developed as a result.

Dissertation Research Diary:

Supervision Meeting 1: May 26th, 2026 at 4:30 PM

Work completed since last meeting:

Submitted the dissertation proposal and prepared questions on the ethics application process.

Meeting contents:

- Reviewed written feedback on the proposal and discussed how to complete the ethics application.
- Proposal feedback:** the literature review did not make clear which cited studies were machine learning, differential expression, or laboratory functional studies. The literature review should also be expanded to 3,000–4,000 words.
- Research questions, aims, and objectives were confirmed as clear.
- Advised to add a glossary of medical, statistical, and acronym terms, and to add sensitivity and specificity to the performance metrics.
- Noted that the dataset merging process in the methodology was not described in sufficient detail.

Goals before next meeting:

Submit the ethics form for approval, begin the code, and implement the proposal revisions above.

Supervision Meeting 2: June 25th, 2026 at 3:00 PM

Work completed since last meeting:

Submitted the ethics application and made the proposal revisions agreed in Meeting 1.

Meeting contents:

- Confirmed that ethics approval had been granted. Was advised to attach the approval letter as an appendix.
- Agreed on a working timeline, including a full draft to be sent for review by 17th August ahead of submission.

Goals before next meeting:

Complete the code and write a substantial portion of the dissertation for discussion.

Supervision Meeting 3: July 21st, 2026 at 11:15 AM

Work completed since last meeting:

Progressed the code. Writing was delayed by methodological issues that arose during implementation.

Meeting contents:

- Discussed how many genes the elastic net should retain before model training (384 versus 64), and how to justify the choice.
- Discussed the classifiers all achieving perfect performance on the dataset and how to justify this.
- Discussed refining the research questions so the study offered a clearer contribution.

Goals before next meeting:

Submit the full draft to Dr. Bath for discussion at the next meeting.

Supervision Meeting 4: August 20th, 2026 at 2:00 PM

Work completed since last meeting:

Sent the first three chapters for review, with two chapters remaining before the deadline.

Meeting contents:

- Advised that the draft was on the right track and needed only minor adjustments.
- Gave formatting guidance: using Word's "keep with next" to keep captions with figures, relabelling chapter headers as "Chapter 1" etc., numbering figures by chapter (e.g. Figure 3.1), and reducing table font size to 10 pt.
- Suggested minor wording changes, including a typo in Research Objective 1.
- Approved putting large result files and the code file on Google Drive, referenced in the dissertation, rather than as appendices.

Goals before next meeting:

This was the final supervision meeting before submission.

Reflections on Research:

The following five reflections each discuss the feedback received, the action taken and its justification, and the skill developed. Each reflection was first recorded on mySkills.

Supervision Feedback 1:

Reflection on Feedback:

Dr. Bath noted on my proposal feedback that my proposal's literature review did not make clear which cited studies were machine learning, differential expression, or laboratory functional work, so the reader could not easily see how these sources related to each other. In response, I reorganised the review into clearer sections and added framing that made it clear which methodology type each section drew on. I judged this more effective than the proposal's combined structure because differential expression shows which genes differ, functional studies show what those genes do, and machine learning asks which genes a predictive model relies on. Grouping the literature by methodology type stops these distinct forms of evidence being read as one and the same.

Associated skill: Research and critical thinking

I learned to synthesise a body of literature by methodology rather than summarise it study by study, which sharpened my ability to read a field critically and see what is established and where the gaps lie.

Supervision Feedback 2:

Reflection on Feedback:

Dr. Bath advised on my proposal feedback that I should add sensitivity and specificity as performance metrics, which capture false negatives and false positives directly. I added both to my selected performance metrics and explained what each captures clinically. These matter in a diagnostic setting because a missed psoriasis case and a patient without psoriasis wrongly flagged carry different consequences for the patient, so measuring each separately gives a clearer picture of how the model would perform in practice.

Associated skill: Applying knowledge

I learned that performance metric choice is not a default but a decision that must reflect the real-world costs of error, and I now approach classification evaluation by asking what a mistake would mean in context rather than reaching for standard measures.

Supervision Feedback 3:

Reflection on Feedback:

In meeting 3, I was uncertain whether the elastic net feature selection should narrow the panel to 384 or 64 genes before model training. Dr. Bath advised that either was defensible provided I could justify the choice. I chose 64 to produce a more compact and interpretable panel, justified by showing that reducing the panel from 384 to 64 genes increased cross-validation error by only 0.003, a negligible cost for a substantially smaller and more interpretable set. Interpretability was central to my aim, as a smaller panel makes the downstream SHAP analysis and candidate shortlist easier to examine and interpret.

Associated skill: Research and critical thinking

I learned to resolve a methodological choice by weighing it against my own research aims and the evidence rather than defaulting to the lower-error option, which has made me more confident making independent design decisions and defending them.

Supervision Feedback 4:

Reflection on Feedback:

Also in meeting 3, I raised a concern that all three classifiers achieved perfect performance on the dataset. Dr. Bath advised that this was acceptable if evidenced. I addressed it in two ways. First, I cited studies using the same GSE54456 dataset that reported the same perfect separability, showing the result reflected the strength of the psoriatic signal rather than an error. Second, and more importantly, I recognised that if classification was guaranteed, performance metrics alone offered little contribution, so I sharpened the research gap and questions accordingly. RQ2 was reframed from whether models could predict psoriasis to how far independent algorithms agree on the most important genes, and RQ3 around whether genes are supported as robust biomarkers through convergence of three evidence lines. This shifted the study's gap and contribution from classification, which was already established, to identifying which genes are robustly supported across classification models and evidence lines.

Associated skill: Research and critical thinking

I learned to treat an unexpected result as a prompt to question my own assumptions, and to reframe research questions so a study contributes something new rather than confirming what is already known, a way of thinking I expect to rely on in future research.

Supervision Feedback 5:

Reflection on Feedback:

Dr. Bath recommended in meeting 4 that I include a glossary listing the biological, statistical, and machine learning terms and acronyms used in the dissertation, each with a brief description. Therefore, I compiled a glossary covering both the biological vocabulary (such as differential expression and pleiotropy) and the machine learning vocabulary (such as SHAP, elastic net, and nested cross-validation). I judged this as important because my dissertation is interdisciplinary, and a reader strong in one field may not be fluent in the other.

Associated skill: Interpersonal skills

I learned to anticipate a reader who may know one half of an interdisciplinary field but not the other, and to make specialist content accessible without diluting it. This is a communication skill that I now recognise as essential whenever I write across disciplines.

Screenshot of MySkills:

The screenshot displays the 'mySkills' interface of the University of Sheffield's Careers & Employability Service. The header includes the university logo and navigation links for Search, Book, mySkills, My Career Passport, and Career Discovery. The main section is titled 'mySkills' and features a tabbed interface with 'My development experiences' selected. Below this, there are five feedback entries, each with a date of 23.08.2026 and a skill tag. The entries are:

- Supervision Feedback 1:** Feedback on separating study types in the literature review. Skill: RESEARCH AND CRITICAL THINKING.
- Supervision Feedback 2:** Feedback on adding sensitivity and specificity performance metrics. Skill: APPLYING KNOWLEDGE.
- Supervision Feedback 3:** Feedback on elastic net feature selection. Skill: RESEARCH AND CRITICAL THINKING.
- Supervision Feedback 4:** Feedback on perfect classification performance. Skill: RESEARCH AND CRITICAL THINKING.
- Supervision Feedback 5:** Feedback on including a glossary. Skill: INTERPERSONAL SKILLS.

At the top right of the feedback section, there are buttons for 'Export' and '+ Add new'.

Appendix G: AI Self-Declaration

I acknowledge the use of AI software to generate the nested cross-validation figure (Figure 3.6), to check spelling and grammar, and, on occasion, to refine the phrasing of my own writing. This dissertation is a genuine account of the research I have undertaken, and the content can still be considered my own, with all references cited accordingly.

The following details are provided in accordance with the above declaration:

Name and version of the generative AI system used: Claude Opus 4.8.

Publisher: Anthropic.

URL of the AI system: <https://claude.ai>

Purpose and use of the AI output: The tool was used for the three limited purposes set out below. All output was critically reviewed, fact-checked, and edited by me, and I remain the primary author of and am fully accountable for this dissertation.

First, the nested cross-validation design in Figure 3.6 (Section 3.5.4) was drafted with the tool. I described the design of the outer and inner folds in words and used the resulting draft as a starting point, then corrected and adapted it so that it accurately reflected the design implemented in this study. Second, the tool was used to check spelling and grammar across the thesis. Third, it was used on occasion to suggest alternative phrasings for sentences that I had already written. I retained full judgement over which suggestions, if any, to adopt.

Representative prompts used included:

-“Create a diagram of my nested cross-validation design with five stratified outer folds and an inner five-fold tuning loop, with feature selection applied once to each outer-training set.”

-“Check the following text for spelling and grammatical errors and suggest corrections without changing my meaning.”

-“Suggest a more academic phrasing for this sentence while keeping the meaning the same.”